Week 5 · Waypoint Following and LOS Guidance

USV Guidance, Navigation and Control (Graduate) · Department of Autonomous Vehicle System Engineering, Chungnam National University | revised 2026-09-22

ℹ Recordings of this semester's own classes

Every session of this course is recorded as it is given and posted to the playlist below. The written material here is the reference; the recording is the lecture, at the pace it was actually delivered. Use it to review a derivation, or to catch up on a missed week before the laboratory.

	Where
[충남대] 2026 센서신호처리 — this course, recorded live, 2026	playlist

★ Reference material — read this first

The five courses below are **taught by the instructor of this course** and are the assumed background for it. They run from the fundamentals down to the graduate material, so start wherever the gap is. Every frame, symbol and derivation used here is developed in them at length; anyone whose prerequisites are thin should work through them first, then return.

#	Course	Level	Lang.	Video	Slides and code
1	**Control Engineering** (제어공학) — the foundation: transfer functions, feedback, stability, root locus, PID	undergraduate	KO	playlist	drive
2	**Control System Design** (제어시스템설계) — design rather than analysis: specifications, loop shaping, discrete implementation	undergraduate	KO	playlist	drive
3	**Advanced Control Engineering** (제어공학특론) — reference frames, the 6-DOF equation of motion, rotation matrices and Euler angles, linearisation and trim, vehicle control design	graduate	EN	playlist	drive
4	**Sensor Signal Processing and Fusion** (센서신호처리 및 융합) — sensor models, noise, estimation and multi-sensor fusion	graduate	EN	playlist	drive
5	**Capstone Design** (캡스톤디자인) — a vehicle project carried end to end	undergraduate	KO	playlist	drive

Not fluent in MATLAB or Simulink yet? Do these before Part 2. Every laboratory in this course is Simulink, and the Onramp courses are free and take a few hours each.

Tool	Where to start
MATLAB	MATLAB Onramp · Core MATLAB Skills
Simulink	Simulink Onramp · instructor's Simulink lectures part 1 · part 2

✎ Getting the course files, and keeping them current (Windows)

The notes, models and scripts are kept in one Git repository that is updated through the semester. Clone it once; before every class, pull. A pull downloads only what has changed since the last one.

When	Where to run it	Command
once	PowerShell — installs Git for Windows	`winget install --id Git.Git -e`
once	the folder that will hold the course, e.g. `Documents`	`git clone https://github.com/wkyouncnu/Sensor-Signal-Processing-and-Fusion.git`
before every class	inside the cloned folder `Sensor-Signal-Processing-and-Fusion`	`git pull`

- `git pull` prints `Already up to date.` when nothing has changed, and otherwise lists the files it updated.
- The repository is private. When Git asks, sign in with the GitHub account the instructor has given access.
- Experiment on copies, not on the cloned files: copy a week's `WXX_simulink` folder elsewhere first, and a pull can never collide with local edits. If it already has, `git stash`, then `git pull`, then `git stash pop` sets the edits aside, updates, and puts them back.
- The MSS toolbox is not part of the repository. The weeks that simulate the Otter need it at `Tools\MSS` inside the cloned folder.

- **Course**: USV Guidance, Navigation and Control (Graduate)
- **Department**: Autonomous Vehicle System Engineering, Chungnam National University
- **This week**: ① where the heading command comes from: guidance turns a list of waypoints into ψ_d ② line of sight (LOS) as a P controller on the cross-track error, tuned by its look-ahead distance ③ when to move to the next leg ④ a current, and the integral that removes the offset it leaves (ILOS) — all tuned from measurements, with no model of the vessel

★ Prerequisites from the previous week

- Week 4 tuned a heading autopilot on the Otter. This week puts a guidance law in front of it and does not retune it: the autopilot follows whatever heading it is given.
- Week 4 §4-4: heading is not course. A vessel in a current points one way and travels another; §5-5 is where that difference costs a path error.
- From Weeks 2 to 4: P speeds a loop up and leaves an error when something pushes back; I removes that error; the tuning order.

Learning Outcomes

Upon completion of this week, the learner is able to:

1. Compute the cross-track error of a vessel from the two waypoints of a leg, and explain why aiming at a waypoint does not follow the line between waypoints.
2. Write the LOS law, recognise it as a P controller on the cross-track error with gain $1/\Delta$, and choose Δ from a measured sweep.
3. Choose the switching distance from the corner behaviour it produces on a mission.
4. Predict the offset a cross current leaves under LOS, $\Delta \tan$ of the heading held, and remove it with ILOS.
5. Select a guidance law with one flag, and compare the three on one mission with everything else held fixed.

6. Tune a guidance law by the order $\Delta \to R \to \kappa$, stating the measurement behind each choice.
7. Derive the allocation of a surge force and a yaw moment onto two fixed propellers, and state the moment limit it leaves as a function of the surge force.
8. Run the speed loop of Week 3 and the guidance of Week 5 on one hull, and account for what the combination costs.

Prerequisites and Setup

Item	Requirement
MATLAB	R2024b, with Simulink
MSS	`Tools/MSS`, found by `W05_0_setup` through `mss_path`
Course folder	`lectures/W05_simulink`
Models	**one per experiment** — `W05_C_atan2`, `W05_D_LOS`, `W05_E_switching`, `W05_F_LOS`, `W05_F_ILOS`, `W05_G_tuning` — all generated by `W05_1_build_guidance`, never edited by hand
Time	lecture 40 min, laboratory 1 h 20 min

Part 1 · Principle and experiment, section by section

Every section states what is observed, says what follows from it, and then runs **the experiment that measures it**, on its own model. The numbers quoted in the text are printed by the run a few lines below it.

Part of a section	What it holds
What is observed	the effect itself, with the numbers it produces
What follows from it	numbered lines, one step each, from the geometry of the path to the law
Experiment N-x	the model, the commands, the output actually obtained, the figure, and a table that reads every feature of the figure back to **the numbered line that predicts it**

5-0. Setting up (10 min)

```
cd lectures/W05_simulink
W05_0_setup
W05_1_build_guidance
```

Expected output:

```
  W05_0_setup
    path      5 waypoints, legs of 60 m
    guidance  Delta = 5 m (P gain 0.200), R_switch = 3 m, kappa = 0.3
    start     20 m east of the path;  current 0 m/s

  built  W05_C_atan2.slx      (overlapping lines: 0)
  built  W05_D_LOS.slx        (overlapping lines: 0)
  built  W05_E_switching.slx  (overlapping lines: 0)
  built  W05_F_LOS.slx        (overlapping lines: 0)
  built  W05_F_ILOS.slx       (overlapping lines: 0)
  built  W05_G_tuning.slx     (overlapping lines: 0)
```

- `W05_0_setup.m` is the only file edited by hand. If it stops in `mss_path`, the MSS toolbox is not at `Tools\MSS`.
- The first line of `W05_0_setup` is `clear`. Run it again whenever a Scope does not match these notes.

5-1. What guidance adds

This section answers: where does the heading command come from?

- Weeks 3 and 4 were given their commands by hand: a speed, a heading. A vessel on a mission is given **waypoints** instead, and something must turn them into a heading to steer. That something is **guidance**.
- Guidance is a controller around the heading autopilot. Its output, ψ_d, is the autopilot's input; its input is the vessel's position; its error is the distance from the path.

Every model this week has the same four blocks. Each is a MATLAB Function with commented code; double-clicking one shows the equations of this section.

Block	Input → output	This week
`guidance`	position → commanded heading ψ_d	the subject: atan2, LOS or ILOS
`heading autopilot`	ψ_d and the state → yaw moment N	the gains of Week 4, $K_p = 300$, $K_d = 100$, with P on the wrapped heading error and D on the measured yaw rate — so that the jump in ψ_d at a waypoint switch gives no derivative kick (Week 2 §2-10); not retuned
`allocation`	N → two shaft speeds	Week 4 §4-1, with a constant surge force $X_{ff} = 60$ N (about 0.77 m/s)
`Otter`	shaft speeds → twelve states	Week 1

The error guidance works on. "Follow the path" has to become a number before a controller can act on it, and **one number is not enough**. The path is a straight leg from waypoint k to waypoint $k{+}1$; what is needed is how far **along** it the vessel has come and how far **off** it the vessel is. Both come out of one construction.

Step 1 — the leg gives the frame its direction. The two waypoints fix a direction measured from north:

$$\pi_p = \operatorname{atan2}\left(E_{k+1} - E_k,\; N_{k+1} - N_k\right)$$

- π_p is **constant along a leg** and is recomputed only when the waypoint index changes.
- It must be the two-argument `atan2`. A leg can run into any of the four quadrants, and the one-argument $\arctan$ folds two of them onto the other two — the same reason Week 4 §4-6 needs it.

Step 2 — the leg carries two unit vectors. Along the leg, and to its right:

$$\hat{\mathbf{t}} = \begin{bmatrix} \cos\pi_p \\ \sin\pi_p \end{bmatrix} \qquad \hat{\mathbf{n}} = \begin{bmatrix} -\sin\pi_p \\ \cos\pi_p \end{bmatrix}$$

They are perpendicular and of unit length, so together they are a **frame** — the path frame $\{p\}$ — and $\hat{\mathbf{n}}$ points to starboard of a vessel running along the leg.

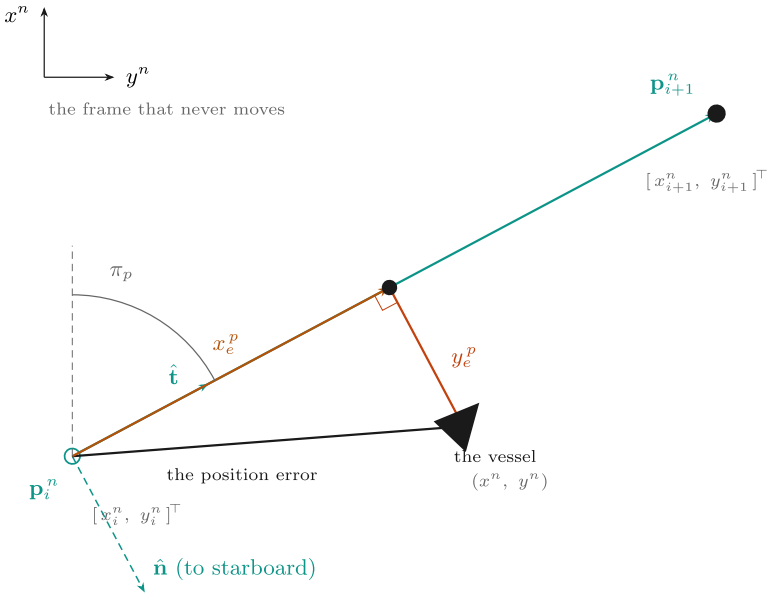
Step 3 — resolve the position error in that frame. Take the vector from the waypoint to the vessel and ask for its two components:

$$\begin{bmatrix} x_e \\ y_e \end{bmatrix} = \mathbf{R}(\pi_p)^{\mathsf{T}} \begin{bmatrix} N - N_k \\ E - E_k \end{bmatrix}, \qquad \mathbf{R}(\pi_p) = \begin{bmatrix} \cos\pi_p & -\sin\pi_p \\ \sin\pi_p & \cos\pi_p \end{bmatrix}$$

which written out is the pair this week uses everywhere:

$$\begin{aligned} x_e &= \;\;(N - N_k)\cos\pi_p + (E - E_k)\sin\pi_p && \text{along the leg} \\ y_e &= -(N - N_k)\sin\pi_p + (E - E_k)\cos\pi_p && \text{across it, positive to starboard} \end{aligned}$$

Symbol	Quantity	Unit · source
(N_k, E_k), (N_{k+1}, E_{k+1})	the two waypoints of the active leg	m · `WP_N`, `WP_E`
(N, E)	the vessel's position, north and east	m · states 7 and 8
π_p	the direction of the leg, from north	rad
x_e	**along-track error**: distance travelled along the leg	m · §5-4 switches on it
y_e	**cross-track error**: distance off the leg, positive to its right	m · §5-3 drives it to zero
$d = \lVert\mathbf{p}_{k+1} - \mathbf{p}_k\rVert$	the leg's length	m · $d - x_e$ is what is left to run



one rotation, two answers

Known: the two waypoints $\mathbf{p}_i^n = [\,x_i^n,\ y_i^n\,]^\top$ and $\mathbf{p}_{i+1}^n = [\,x_{i+1}^n,\ y_{i+1}^n\,]^\top$, and the vessel's position $(x^n,\ y^n)$.

The leg gives the frame its direction:

$$\pi_p = \operatorname{atan2}(y_{i+1}^n - y_i^n,\ x_{i+1}^n - x_i^n)$$

and the error, written in that frame, is

$$\begin{bmatrix} x_e^p \\ y_e^p \end{bmatrix} = \mathbf{R}_p^n(\pi_p)^\top \begin{bmatrix} x^n - x_i^n \\ y^n - y_i^n \end{bmatrix}$$

x_e^p — how far **along** the leg the vessel has come. The switching test of §5-4 uses it.

y_e^p — how far **to the side**. The guidance law drives this one to zero.

Both are projections of the same vector onto the same orthonormal pair, so $x_e^2 + y_e^2 = \|\mathbf{p} - \mathbf{p}_i^n\|^2$ — a rotation preserves length, and that is the cheapest check on the whole construction.

Reading the figure.

Where to look	What is there	Why
the two black dots on the leg	$\mathbf{p}_i^n = [\,x_i^n,\ y_i^n\,]^\mathsf{T}$ and $\mathbf{p}_{i+1}^n = [\,x_{i+1}^n,\ y_{i+1}^n\,]^\mathsf{T}$	the leg **is** those two numbers; everything else in the figure is computed from them and the vessel's position
the label under the hull	$(x^n,\ y^n)$	the third known: what the navigation reports. The figure of §5-3 uses the same three
the dot where the brown and red segments meet	the foot of the perpendicular	the point the vessel is abeam of: x_e ends there and y_e starts
the dashed pair at waypoint k	$\hat{\mathbf{t}}$ along the leg, $\hat{\mathbf{n}}$ to starboard	step 2: the leg carries its own frame
the black arrow	the position error, before it is resolved	the one vector both errors are components of
the brown segment on the leg	x_e — how far along	step 3, first row
the red segment perpendicular to it	y_e — how far off	step 3, second row
the right angle where they meet	$\hat{\mathbf{t}} \perp \hat{\mathbf{n}}$	which is what makes them independent
the grey arc at the waypoint	π_p, measured from north	step 1

The plainest way to see it: two shadows. The rotation above can be read without any matrix at all, and this is the reading to carry away. Write the vector from the active waypoint to the vessel as

$$\mathbf{d} = \begin{bmatrix} N - N_k \\ E - E_k \end{bmatrix}$$

The leg carries two unit vectors — $\hat{\mathbf{t}}$ along it and $\hat{\mathbf{n}}$ across it — and the dot product of a vector with a unit vector is **the length of its shadow on that direction**. So:

$$x_e = \mathbf{d}\cdot\hat{\mathbf{t}} = \mathbf{d}\cdot\begin{bmatrix}\cos\pi_p\\ \sin\pi_p\end{bmatrix}, \qquad y_e = \mathbf{d}\cdot\hat{\mathbf{n}} = \mathbf{d}\cdot\begin{bmatrix}-\sin\pi_p\\ \cos\pi_p\end{bmatrix}$$

Two dot products, and they are the two lines of code. There is nothing else in the construction: x_e is the shadow on the leg, y_e is the shadow on the perpendicular to it.

A number to hold it down. Take a leg from $(0,0)$ to $(60,60)$ — northeast, so $\pi_p = 45^\circ$ — with the vessel at $(40,0)$, forty metres due north of the first waypoint:

$$x_e = 40\cos 45^\circ = 28.28\ \text{m}, \qquad y_e = -40\sin 45^\circ = -28.28\ \text{m}$$

It is 28.3 m along the leg and 28.3 m to the **port** side of it, which is where a sketch puts it: the leg runs away to the northeast while the vessel sits due north of its start. The negative sign is the convention of property 2 below, not an error.

The same two lines, as one matrix. Stack the two dot products and the rows of the matrix are $\hat{\mathbf{t}}^{\mathsf{T}}$ and $\hat{\mathbf{n}}^{\mathsf{T}}$:

$$\begin{bmatrix}x_e\\ y_e\end{bmatrix}_p = \mathbf{R}_n^p\begin{bmatrix}N-N_k\\ E-E_k\end{bmatrix}_n, \qquad \mathbf{R}_n^p = \begin{bmatrix}\cos\pi_p & \sin\pi_p\\ -\sin\pi_p & \cos\pi_p\end{bmatrix}$$

The matrix is not a new idea; it is the two shadows written together. It is worth having in this form because it is the notation of the literature and of the 2025 course notes, where the frames are named $\{n\}$ for north-east and $\{p\}$ for the path, the waypoint is $(x_i^n,\ y_i^n)$, the leg angle is read as a tangent,

$$\tan\pi_p = \frac{y_{i+1}^n - y_i^n}{x_{i+1}^n - x_i^n}$$

and the second row is the result to remember:

$$\boxed{y_e = -\left(x - x_i^n\right)\sin\pi_p + \left(y - y_i^n\right)\cos\pi_p}$$

with x north and y east. That is the expression above, one symbol at a time.

Where the vessel is abeam of. The foot of the perpendicular — the point **on** the leg that the vessel is level with — follows from the same two vectors with no new work: travel x_e along the leg from the waypoint.

$$\begin{bmatrix}x_p\\ y_p\end{bmatrix}_n = \begin{bmatrix}N_k\\ E_k\end{bmatrix} + x_e\,\hat{\mathbf{t}} \qquad \text{or, as a rotation back,} \qquad \mathbf{R}_p^n\begin{bmatrix}x_e\\ 0\end{bmatrix}_p = \left(\mathbf{R}_n^p\right)^{\mathsf{T}}\begin{bmatrix}x_e\\ 0\end{bmatrix}_p$$

measured from the waypoint. Setting the second component to zero is what "drop a perpendicular" means in algebra: keep the along-track shadow and throw the cross-track one away. The identity $\mathbf{R}_p^n = (\mathbf{R}_n^p)^{\mathsf{T}}$ is the single fact about rotations this week needs — a rotation is undone by its own transpose — and it is why no inverse is ever computed here.

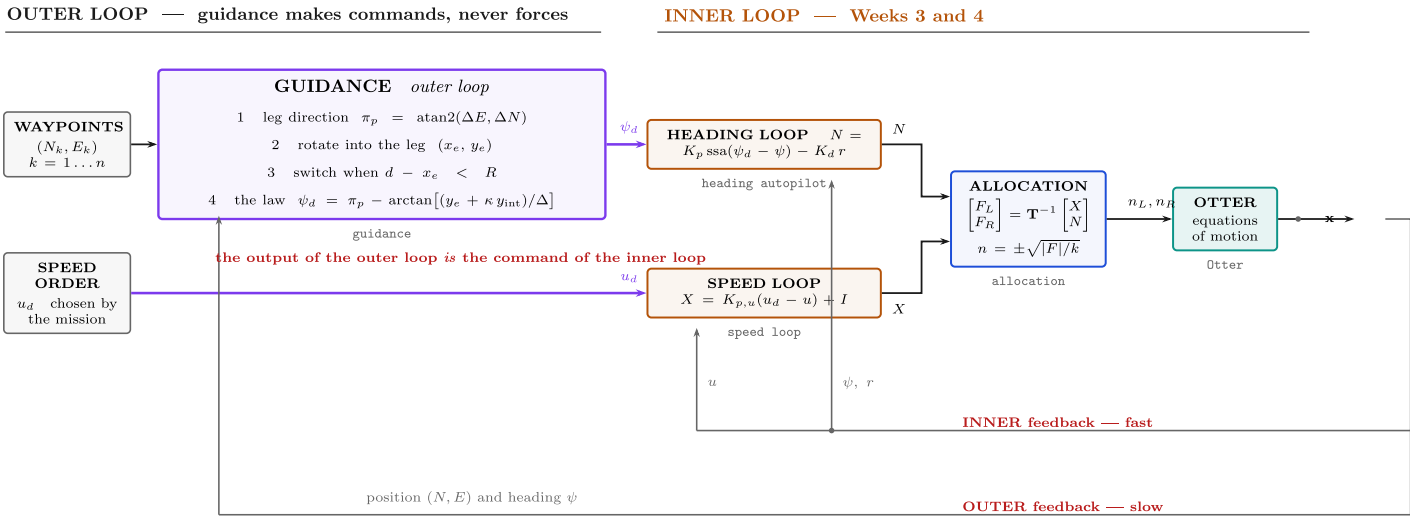
Three properties worth stating before they are used.

1. **It is a rotation, so it preserves length.** $x_e^2 + y_e^2 = \|\mathbf{p} - \mathbf{p}_k\|^2$ exactly. That identity is the cheapest check on the whole construction, and it is what `verify_w05_guidance` check 5 tests.
2. **The sign of y_e is a convention with consequences.** Written as above, $y_e > 0$ means the vessel is to **starboard** of the leg, and §5-3's law then subtracts a positive correction — turning to port, back towards the path. Swap the two sine terms and the vessel runs away from the path at the same speed.
3. **Do not compute y_e from a distance formula.** The perpendicular distance $|(N-N_k)\sin\pi_p - (E-E_k)\cos\pi_p|$ gives the magnitude but throws away the sign and gives no x_e at all — and §5-4 needs x_e. One rotation answers both questions and cannot make them disagree.

> **The same two lines, in Fossen's notation**
>
> This is the form used throughout the field, and the reference implementation is Fossen's TTK4190 lecture code — the source of the version taught in *2. USV 제어기 설계*:
>
> ```
> pi_p = atan2(yk_next-yk, xk_next-xk);        % path-tangential angle w.r.t. North
> % along-track and cross-track errors (x_e, y_e) expressed in NED
> x_e =  (x-xk) * cos(pi_p) + (y-yk) * sin(pi_p);
> y_e = -(x-xk) * sin(pi_p) + (y-yk) * cos(pi_p);
> ```
>
> with (x, y) north and east. The `guidance` block of this week is these lines, and §5-4's switching test is the next two. The rotation gives the same y_e as MSS `crosstrackWpt.m` to machine precision (`verify_w05_guidance` check 5).

- The goal of path following is $y_e \to 0$; x_e is not driven anywhere, it is only **watched**, and §5-4 is what watches it.

Where it attaches. Weeks 3 and 4 built two loops that each close on a measurement the vessel already carries: the speed loop reads u, the heading loop reads ψ and r. Neither of them knows what a path is, and neither needs to. Guidance does not join them — it sits **in front** of them and supplies what they were previously given by hand:



In the diagram	What it is	Which week
`guidance`, the four numbered steps	leg direction, the rotation into (x_e, y_e), the switching test, the law	**this week**
the thick arrow marked ψ_d	the output of the outer loop, and **the command of the heading loop**	this week → Week 4
the thick arrow marked u_d	the speed order, which goes straight past the guidance into the speed loop	§5-7
`heading autopilot`, `speed loop`	unchanged from Weeks 4 and 3, and not retuned	Weeks 3, 4
`allocation`	the two steps of Week 3 §3-6, with both inputs now live	Week 3
INNER feedback, marked fast	u, ψ, r — read every solver step, and the loops are tuned for that	
OUTER feedback, marked slow	position (N, E) and heading, which is what a path is defined against	

- The commands are **drawn in a different colour and a heavier line** for one reason: ψ_d is not a force, a moment, or a shaft speed. It is an *instruction to another controller*, and that is the only kind of thing guidance ever

produces.

- The two feedbacks are nested, not parallel. The inner loop settles a heading in a few seconds; the outer loop changes what heading to ask for over tens of metres. That separation of time scales is what makes the cascade work, and §5-3's choice of Δ is where it is paid for.

> **Nothing in Weeks 3 or 4 changes**
>
> The `heading autopilot` block in `W05_G_tuning` is the block of Week 4, with the gains of Week 4, unretuned. The only difference is where ψ_d comes in from. If a guidance law cannot be added without retuning the autopilot beneath it, the two were never properly separated.

Experiment 5-1 · The guidance layer on the canvas (10 min)

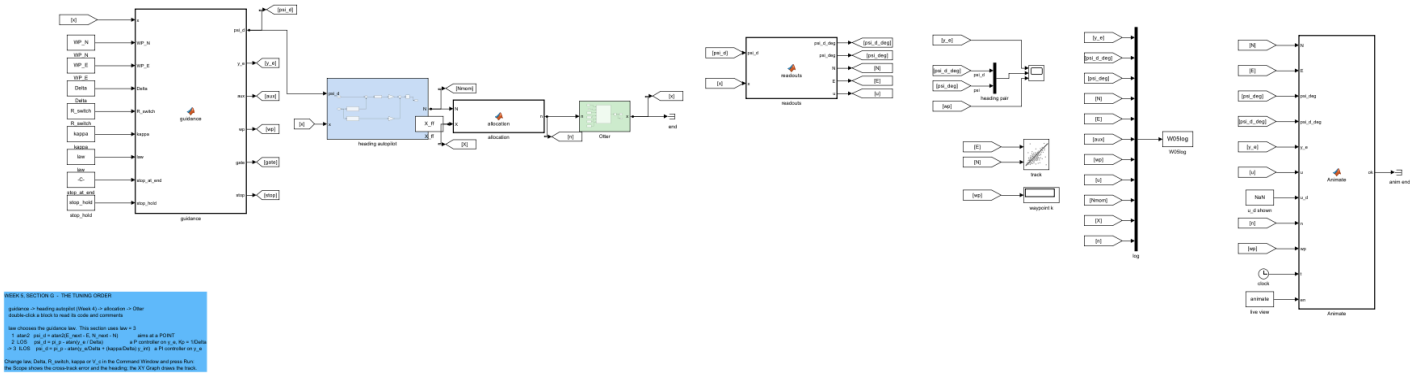
What it measures. Nothing yet: the four blocks of the table above are matched to the canvas, and the rotation that produces y_e is read inside the block that performs it.

The canvas. `W05_G_tuning` — the fullest model of the week, read here and measured in Experiment 5-6. Everything except the `guidance` block is Week 4, unchanged and not retuned.

Opening and running.

```
W05_0_setup
open_system('W05_G_tuning')        % Run — XY 그래프에 항적이 그려진다 / the track is drawn live
```

- Double-click `guidance` and read the numbered comments: the active leg, the rotation into (x_e, y_e), the switching test of §5-4, and the law of §5-3 and §5-5.
- Double-click `heading autopilot`: the two gains of Week 4 and nothing else. The path does not appear anywhere in it.



In the figure	Meaning
`guidance`	the guidance law of §5-3 and §5-5; outputs ψ_d, y_e, the integral `aux` and the leg number `wp`
`heading autopilot`	the Week 4 gains, D on the yaw rate (§5-1), not retuned
`allocation` , `Otter`	the thrust split of Week 4 and the vessel
Goto `[x]` after the Otter	the twelve states, sent to the guidance, the autopilot and the readouts
`readouts`	the headings in degrees and the position, for display only
Scope, `track` (XY Graph), `W05log`	the cross-track error and headings; the track, east against north; the log

> ✎ **In class**
>
> - **Purpose** — show where guidance sits: one block in front of the autopilot of Week 4, which is not touched.
> - **Point to** — double-click `guidance` and read the numbered comments: the leg, the rotation to y_e, the switching test, the law.
> - **Ask** — "What does the autopilot know about the path?" Nothing. It follows ψ_d; the path lives only in the guidance.
> - **Take away** — guidance is a controller whose output is another controller's command.

Experiment 5-1b · One model, three laws (10 min)

The `guidance` block does not hold one law. It holds **four**, and one input chooses between them:

`law`	Law	What it aims at	Derived in
1	atan2	the next waypoint — a **point**	§5-2
2	LOS	a point Δ ahead on the **line**	§5-3
3	ILOS	the same line, while remembering the push	§5-5
4	ILOS, plain integral	the same line and the same aiming expression, with the textbook integrator	§5-5 line 9b

Laws 3 and 4 differ in exactly one line of code — how the integral state is updated — so §5-5c can attribute every difference between them to that line.

The value arrives from the Constant named `law` on the canvas, which reads the workspace variable of that name. Changing it and pressing Run makes the same model follow a different law, with no rebuild:

```
W05_0_setup
law = 1;                          % 또는 2, 3, 4 / or 2, 3, 4
open_system('W05_G_tuning')       % Run
```

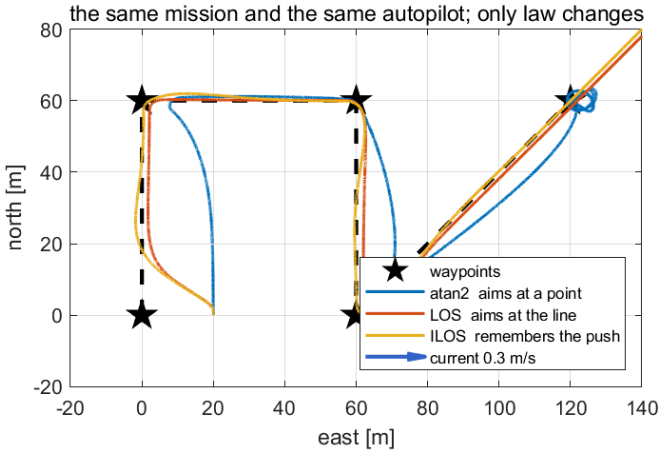
Because everything else is held fixed — the same waypoints, the same current, the same Week 4 autopilot, the same Δ and R — the difference between the three runs is the law and nothing else. That is what makes the comparison worth anything.

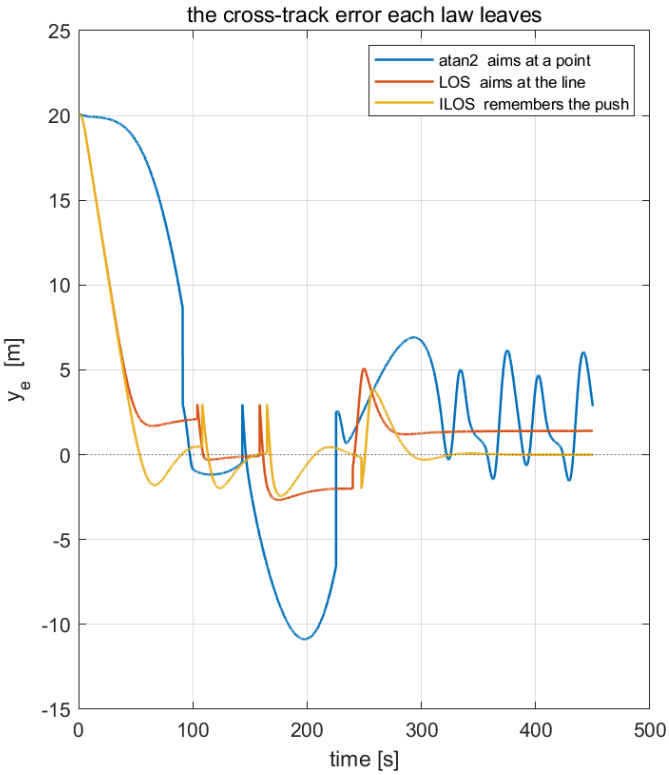
To produce every figure and number in this section

```
cd lectures/W05_simulink
W05_0_setup
W05_B_three_laws
```

Actual output:

```
W05 Experiment 5-1b  one model, three laws  (W05_G_tuning, current 0.3 m/s east)
  law   mean |y_e| legs 2-4 [m]   worst |y_e| [m]   last waypoint at [s]
  atan2                3.929            10.877                316.1
  LOS                  1.521             5.102                330.4
  ILOS                 0.614             3.847                338.8
```





In the figure	Meaning
left, the three tracks	the same waypoints and the same autopilot; only `law` differs
left, the blue loop at the last waypoint	atan2 orbiting the point it is aiming at
right	the cross-track error each law leaves, on the same axes
the arrow	the current, 0.3 m/s towards the east

Reading the figure and the table.

Where to look	What is there	What it says
the blue track, bulging off every leg	3.93 m mean, 10.88 m worst	aiming at a point does not follow the line between points — §5-2 makes this exact
the blue loop at $(120, 60)$	it circles the last waypoint	a point gives no direction once reached; a line still does
the orange track, parallel to each leg	1.52 m mean, and **steady**	LOS holds the line's direction but sits beside it: the offset a current leaves — §5-5 predicts its size
the yellow track, on the legs	0.61 m mean	the integral takes over the push — §5-5b
the right panel after $t = 300$ s	blue swinging, orange flat and offset, yellow on zero	three different failures and one success, on one pair of axes
the last column	$316 \to 330 \to 339$ s	each step costs time. Holding a line means not cutting corners, and not cutting corners is further to travel

What the figure says

- The three laws are not three tunings of one idea; they aim at different things, and the table is the order in which a guidance law grows.

What to try	What to watch
`law = 1; V_c = 0;` Run	with the current removed atan2 still misses the legs: 1.69 m mean and 4.44 m at the worst. The error is geometric, not a disturbance, and that is the whole of §5-2
`law = 2; kappa = 3;` Run	the track does not move by one bit — κ belongs to ILOS, and law 2 never reads it. The block shows why: `y_int` is set to zero on that branch
`law = 3; Delta = 40;` Run	the integral still removes the offset, but the approach takes 118 s (§5-3). Δ and κ do different jobs

> ✎ **In class**
>
> - **Purpose** — show the three laws as one question asked three ways, before deriving any of them.
> - **Point to** — the blue loop at the last waypoint. Ask what a vessel aiming at a point it has already reached is supposed to do.
> - **Ask** — "Why is the fastest run also the worst?" Cutting a corner is shorter. Path following is the decision not to.
> - **Take away** — one block, one switch, everything else held fixed: that is what makes three runs a comparison instead of three anecdotes.

5-2. Aiming at a point is not following a line

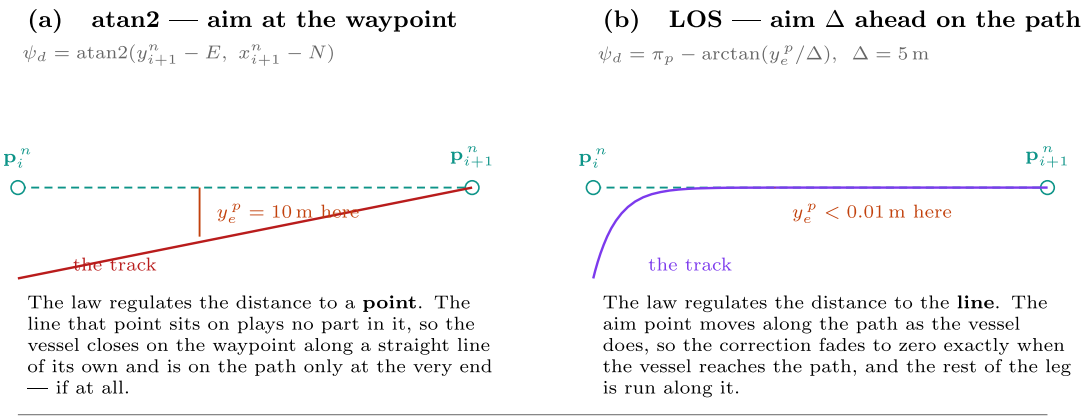
This section answers: why not simply point at the next waypoint?

What is observed. The vessel starts 20 m east of a path running north and is steered by the obvious law, $\psi_d = \operatorname{atan2}(E_{k+1} - E,\ N_{k+1} - N)$ — point at the next waypoint:

- halfway to the waypoint it is still 10.75 m off the path, more than half its starting offset,
- it reaches the path only **at** the waypoint, and then leaves it again on the next leg,
- while the law of §5-3, given the same start, is within 0.04 m of the path at the same point.

What follows from it, one line at a time.

1. `atan2` closes the distance to a **point**. The line it draws is the line from the vessel to the waypoint, and that line is not the path unless the vessel is already on it.
2. The quantity that measures "off the path" is therefore not the distance to the waypoint but the perpendicular distance to the **line**, the cross-track error y_e of §5-1.
3. `atan2` never uses y_e, so nothing in it drives y_e to zero: the error falls only as a side effect of arriving, which is why it vanishes exactly at the waypoint and not before.
4. A guidance law that follows a path must take y_e as its input and return a heading that reduces it. That is §5-3, and it makes the law a feedback controller like every other one in this course.
5. On a real mission the waypoints are far apart and the space **between** them is where the vessel is asked to be, so the error of line 1 is the whole mission rather than an approach transient.



Both vessels start 20 m off the path and both reach the waypoint. Only one travels along the line to it. The tracks are not sketched: the LOS curve is the guidance law integrated along the path, $\mathrm{d}y^p_e/\mathrm{d}s = -y^p_e/\sqrt{\Delta^2 + y^2_e}$, by `_tools/w05_track_curves.awk`, so the shape is the law's and not a sketch. The drawn curves are that ideal kinematics; Experiment 5-2 runs the same start through the autopilot and the hull and measures 10.75 m against −0.04 m at the halfway point.

Reading the figure.

Where to look	What is there	Which line predicts it
(a), the red track	a straight line from the start to the waypoint	line 1: `atan2` closes the distance to a **point**, and the shortest way to a point is a straight line
(a), the label at halfway	$y_e = 10$ m, half the starting offset, purely by geometry	line 3: nothing in the law is acting on y_e
(a), where the track meets the path	only at the waypoint itself	line 3: the error vanishes as a side effect of arriving
(b), the purple track	it bends onto the path and then runs along it	line 4: the law takes y_e as its input
(b), the label at halfway	under 0.01 m — the law has already finished	§5-3: the correction fades as the error does
both panels	the same start, the same waypoint, the same vessel	the only difference is which quantity the law reads

- The drawn curves are the guidance law **integrated along the path**, $\mathrm{d}y_e/\mathrm{d}s = -y_e/\sqrt{\Delta^2 + y_e^2}$, not sketches — so the shape is the law's own. Experiment 5-2 puts the same start through the autopilot and the hull, where the vessel lags slightly:

Measured by Experiment 5-2, below:

Law	y_e at 50 m north	at 100 m (the waypoint)	at 150 m
atan2	10.75 m	0.50 m	−0.74 m
LOS (§5-3)	−0.04 m	0.01 m	0.01 m

Experiment 5-2 · Aiming at the waypoint (10 min)

What it measures. Lines 1 and 3: how far from the path the obvious law leaves the vessel, at three points along one leg, against the law that uses y_e.

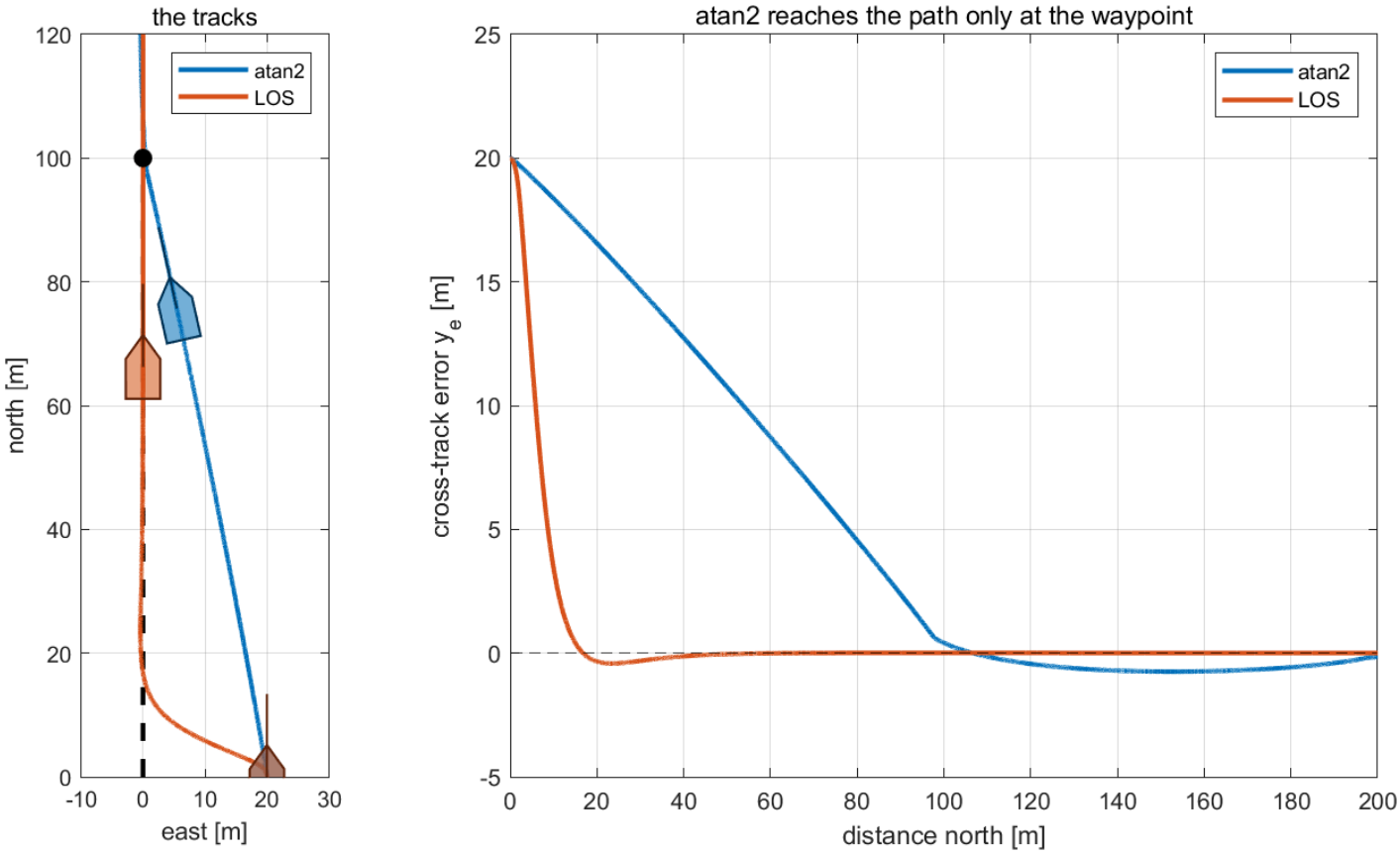
The model. `W05_C_atan2` — the guidance block switched to the `atan2` law, everything behind it (autopilot, allocation, Otter) exactly as Week 4 left it. The waypoints are set to one straight path north, so that the two laws differ in nothing but their own definition.

Opening and running.

```
W05_0_setup
WP_N = [0 100 200 300]'; WP_E = [0 0 0 0]';    % 북쪽으로 곧은 경로 / a straight path north
open_system('W05_C_atan2')          % Run — XY 그래프에서 대각선으로 접근한다 / a diagonal approach
W05_C_aim_at_the_waypoint           % 두 법칙을 나란히 / both laws, side by side
```

Expected output:

```
W05 Experiment 5-2  aiming at the waypoint against line of sight (start 20 m east of the path)
  law     y_e at N = 50 m   N = 100 m   N = 150 m
  atan2            10.75        0.50       -0.74
  LOS              -0.04        0.01        0.01
```



In the figure	Meaning
left	the two tracks with hull silhouettes; the dashed line is the path, the dot the waypoint at 100 m
right	the cross-track error against the distance north

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
left panel, the **straightness** of the atan2 track	a diagonal drawn at the waypoint	line 1: the law aims at a point, and a constant bearing gives a straight run
right panel, the atan2 curve at $N = 100$ m	it touches zero, and only there	line 3: the error vanishes on arrival, not before
right panel, the atan2 curve **past** the waypoint	it leaves zero again, -0.74 m	line 1 on the next leg: a new point, a new diagonal
right panel, the LOS curve	on zero within about 20 m and staying there	line 4: a law that is fed y_e drives y_e
left panel, the hull silhouettes on the atan2 track	pointing at the waypoint throughout	line 1 again: that heading is exactly what the law asks for

What the figure says

- atan2 draws a straight line to the waypoint and meets the path only there; LOS is on the path within 20 m.

What to try	What to watch
`E0 = 5;` Run	2.80 m left at $N = 50$ m instead of 10.75 — **the same fraction** of the starting offset, 0.56 against 0.54. Line 1 is a geometric fault, not a matter of size; LOS leaves 0.02 m in the same run
`WP_N = [0 400]'; WP_E = [0 0]'; E0 = 20;` Run	on one long leg the atan2 track is 10.70 m off at the halfway point and 5.58 m off three-quarters of the way: line 5, the error **is** the mission

> ✎ **In class**
>
> - **Purpose** — show that "go to the waypoint" and "follow the path" are different goals.
> - **Point to** — the blue track cutting diagonally to the waypoint while the orange one turns onto the path at once.
> - **Ask** — "Why is atan2 still 10.75 m off halfway?" It reduces the distance to the waypoint, and the line to the waypoint is not the path.
> - **Take away** — guidance must measure the distance to the line, y_e, and act on it.

5-3. Line of sight: a P controller on the cross-track error

This section answers: what law brings the vessel onto the line, and what is its one tuning knob?

The law. Instead of the waypoint, aim at a point on the path a distance Δ ahead:

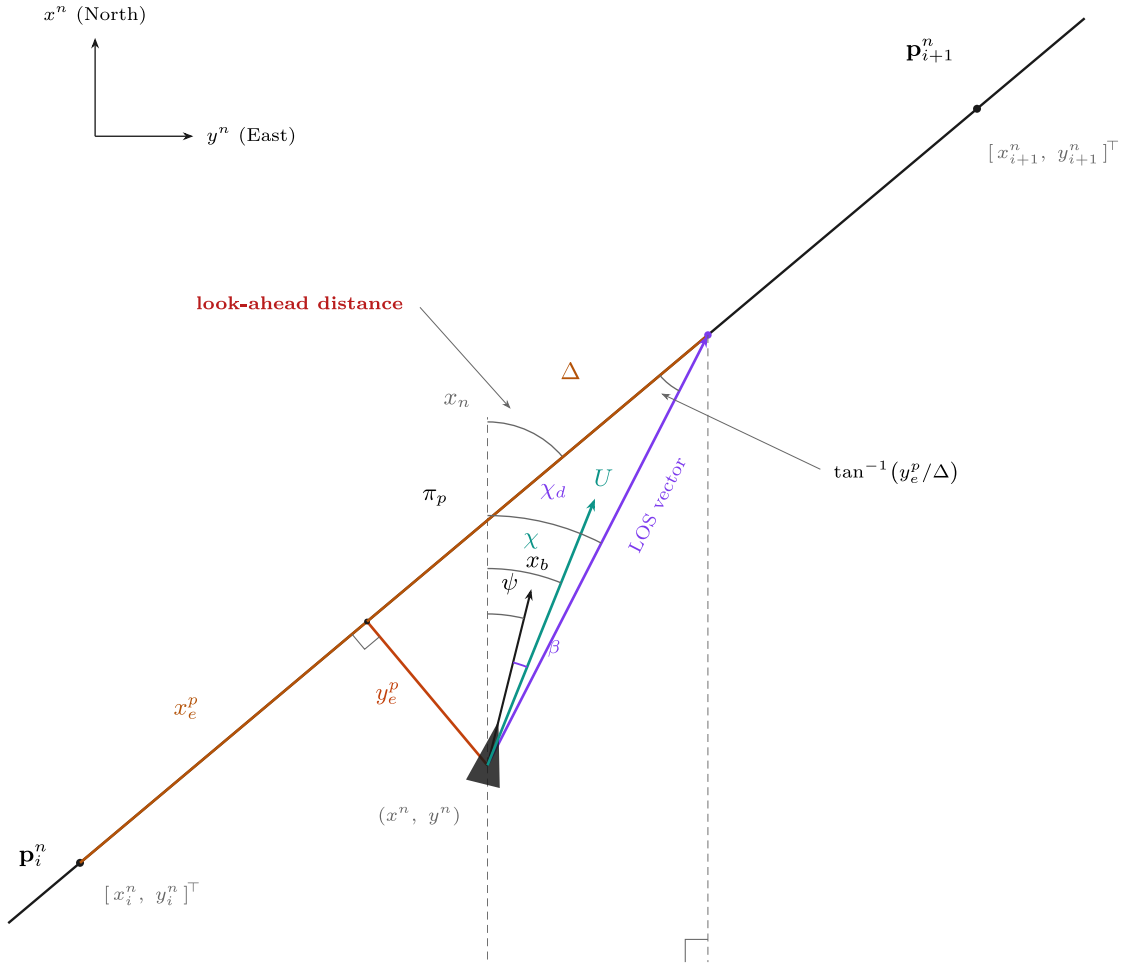
$$\psi_d = \pi_p - \arctan\left(\frac{y_e}{\Delta}\right)$$

- π_p says "line up with the path"; the arctan says "lean towards it, by an angle that grows with how far off the vessel is". On the path, $y_e = 0$ and the vessel simply heads along the leg; far from it, the lean approaches 90° and the vessel heads straight at the path.

Where the arctan comes from. It is not a choice of function — it is the angle of a right triangle. Drop a perpendicular from the vessel to the path; its foot is the point on the path abeam the vessel. Now step a distance Δ **along the path** from that foot: that is the aim point. The triangle with legs Δ (along the path) and y_e (across it) has

$$\tan(\text{the lean}) = \frac{y_e}{\Delta} \qquad \Longrightarrow \qquad \text{the lean} = \arctan\left(\frac{y_e}{\Delta}\right)$$

and the command is $\boldsymbol{\pi_p}$ turned by that lean, towards the path — hence the minus sign, with $\boldsymbol{y_e}$ positive to starboard.



$$\pi_p = \operatorname{atan2}\big(y^n_{i+1} - y^n_i,\ x^n_{i+1} - x^n_i\big)$$

Known

$$y^p_e = -\big(x^n - x^n_i\big)\sin\pi_p + \big(y^n - y^n_i\big)\cos\pi_p$$

Known

$$\mathbf{p}^n_i = [\,x^n_i,\ y^n_i\,]^\top, \quad \mathbf{p}^n_{i+1} = [\,x^n_{i+1},\ y^n_{i+1}\,]^\top$$

Known

$$\chi_d = \pi_p - \tan^{-1}\big(K_p\, y^p_e\big), \qquad K_p = \frac{1}{\Delta}$$

Want to compute

$$y^p_e \to 0, \qquad \chi_d \to \pi_p$$

On the path the correction vanishes and the vessel steers straight along the leg. That is the whole objective, written as a limit.

Two terms, and that is all. π_p lines the vessel up *with* the path; $-\tan^{-1}(y^p_e/\Delta)$ leans it *towards* the path. Nothing else appears, which is exactly why §5-7 can say what the law cannot do.

The correction is bounded by $90°$**.** As $y^p_e \to 0$ the vessel is on the path and $\chi_d \to \pi_p$; as $y^p_e \to \infty$ the law gives $\chi_d \to \pi_p - 90°$, which heads straight at the path and never away from it.

Δ is measured along the path, from the foot of the perpendicular and not from the vessel. The vessel's own distance to the aim point is $\sqrt{\Delta^2 + (y^p_e)^2}$, which is larger.

Notation, and the numbers drawn here

Waypoints are $\mathbf{p}^n_i = [\,x^n_i,\ y^n_i\,]^\top$, resolved in $\{n\}$. The two errors carry the superscript p because they are resolved in the **path** frame $\{p\}$ — not in $\{n\}$ and not in $\{b\}$. The superscripts are what keep the three frames apart.

Drawn with $\pi_p = 50°$, $x^p_e = 32\,\text{m}$, $y^p_e = 16\,\text{m}$ and $\Delta = 38\,\text{m}$, so the correction is $\tan^{-1}(16/38) = 22.83°$ and $\chi_d = 27.17°$; the line-of-sight distance is $41.23\,\text{m}$. The hull is at $\psi = 14°$ with a crab angle $\beta = 8°$, so $\chi = 22°$ — not yet on the commanded course, which is what makes the three arcs at the hull distinct.

Course or heading. The law as written commands a *course* χ_d. This course carries a heading autopilot from Week 4, so §5-3 commands ψ_d: the same expression with ψ_d in place of χ_d. The two agree exactly when $\beta = 0$; Week 4 §4-4 defines the gap and §5-5 measures what it costs on a path.

Reading the figure.

Where to look	What is there	Why
the right angle on the path	the foot of the perpendicular, abeam the vessel	it is the origin the look-ahead is measured from
the segment marked Δ	measured **along the path**, from the foot	not from the vessel: the vessel's own distance to the aim point is $\sqrt{\Delta^2 + y_e^2}$, which is longer
the purple ray	the line of sight, vessel to aim point	the name of the law: the vessel looks at a point and steers at it
the arc marked $\arctan(y_e/\Delta)$	the lean, between the path direction and the line of sight	the right triangle above
π_p at the lower waypoint	the path direction, from north	§5-1 step 1
the two brown segments	x_e and y_e of §5-1	the law uses only the second of them
the three arcs at the hull	ψ, χ and χ_d — heading, course and commanded course	they differ by the crab angle of Week 4 §4-4, which is §5-5's subject

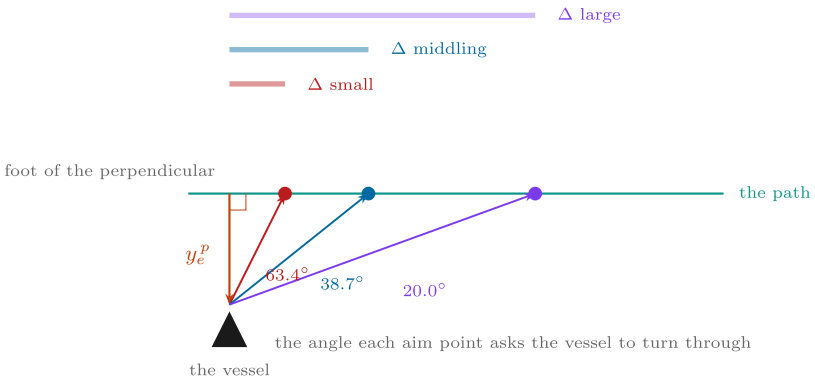
- **Two terms, and that is all.** π_p lines the vessel up *with* the path; $-\arctan(y_e/\Delta)$ leans it *towards* the path. Nothing else appears — which is exactly why §5-5 can say what this law cannot do.
- **The correction is bounded by 90°.** As $y_e \to 0$ the lean goes to zero and $\psi_d \to \pi_p$; as $y_e \to \pm\infty$ it approaches $\mp 90^\circ$, heading straight at the path and never away from it. A law built from a raw gain $-K y_e$ has no such bound and would command absurd headings far from the path.

It is a P controller. For errors small against Δ, $\arctan(y_e/\Delta) \approx y_e/\Delta$ (within 1.4 % for $|y_e| \le 0.2\Delta$, `verify_w05_guidance` check 2), and

$$\psi_d - \pi_p \approx -\frac{1}{\Delta}\, y_e \qquad \Longrightarrow \qquad K_p = \frac{1}{\Delta}$$

Symbol	Quantity	Value
Δ	look-ahead distance	5 m, `W05_0_setup.m`
$1/\Delta$	the P gain: heading correction per metre of error	0.2 rad/m

So Δ is tuned like K_p in Weeks 2 to 4, backwards: a **smaller** Δ is a **larger** gain.



the trade

One and the same y_e^p. Only Δ has changed, and the command has changed by a factor of three.

Δ too small — the arctan saturates almost at once, the vessel turns nearly perpendicular to the path and arrives with speed across it. Measured at $\Delta = 0.5$ m: 1.35 m of overshoot and seven crossings of the path.

Δ too large — the correction is gentle and the vessel closes slowly. At $\Delta = 40$ m it needs 118.3 s to come within 1 m of the path, against 32.4 s at $\Delta = 5$ m.

The rule of thumb is Δ = 2 to 5 hull lengths; for the 2 m Otter that is 4 to 10 m, and this course uses 5 m — two to three lengths. **Experiment 5-3 is the evidence, and the rule of thumb only the sanity check.**

A third cost appears in §5-5: the steady offset a current leaves is $\Delta \tan\beta$, **linear in** Δ.

Reading the figure.

Where to look	What is there	Why
the three aim points on the path	all three lie on the path, at Δ = small, middling, large from the same foot	only Δ has changed
the single red segment y_e	one and the same error in all three cases	so the figure isolates Δ and nothing else
the three angles, 53.1°, 38.7°, 20.0°	what each aim point asks the vessel to turn through	$\arctan(y_e/\Delta)$ with one numerator and three denominators
the ratio between them	a factor of about three in the command, from a factor of three in Δ	Δ is the gain, read backwards

- The trade is the P sweep of Week 2 §2-6 in new clothing: too small a Δ saturates the arctan, turns the vessel nearly perpendicular to the path, and arrives with speed across it; too large a Δ closes gently and slowly.

Measured by Experiment 5-3, below:

Δ [m]	$1/\Delta$	within 1 m after [s]	overshoot [m]	zero crossings
0.5	2.000	27.7	1.35	7
1	1.000	27.9	0.70	2
2.5	0.400	29.1	0.42	2
5	0.200	32.4	0.41	2
10	0.100	41.7	0.41	2
20	0.050	65.2	0.40	1
40	0.025	118.3	0.40	1

- A large Δ is slow; a small one overshoots and swings. $\Delta = 5$ m — two to three hull lengths — reaches the path almost as fast as the smallest values without swinging, and is the choice.

Experiment 5-3 · The look-ahead distance (15 min)

What it measures. That Δ behaves exactly as $1/K_p$: the sweep below is the P sweep of Week 2 §2-6 read backwards, on one long leg so that nothing but the gain is in play.

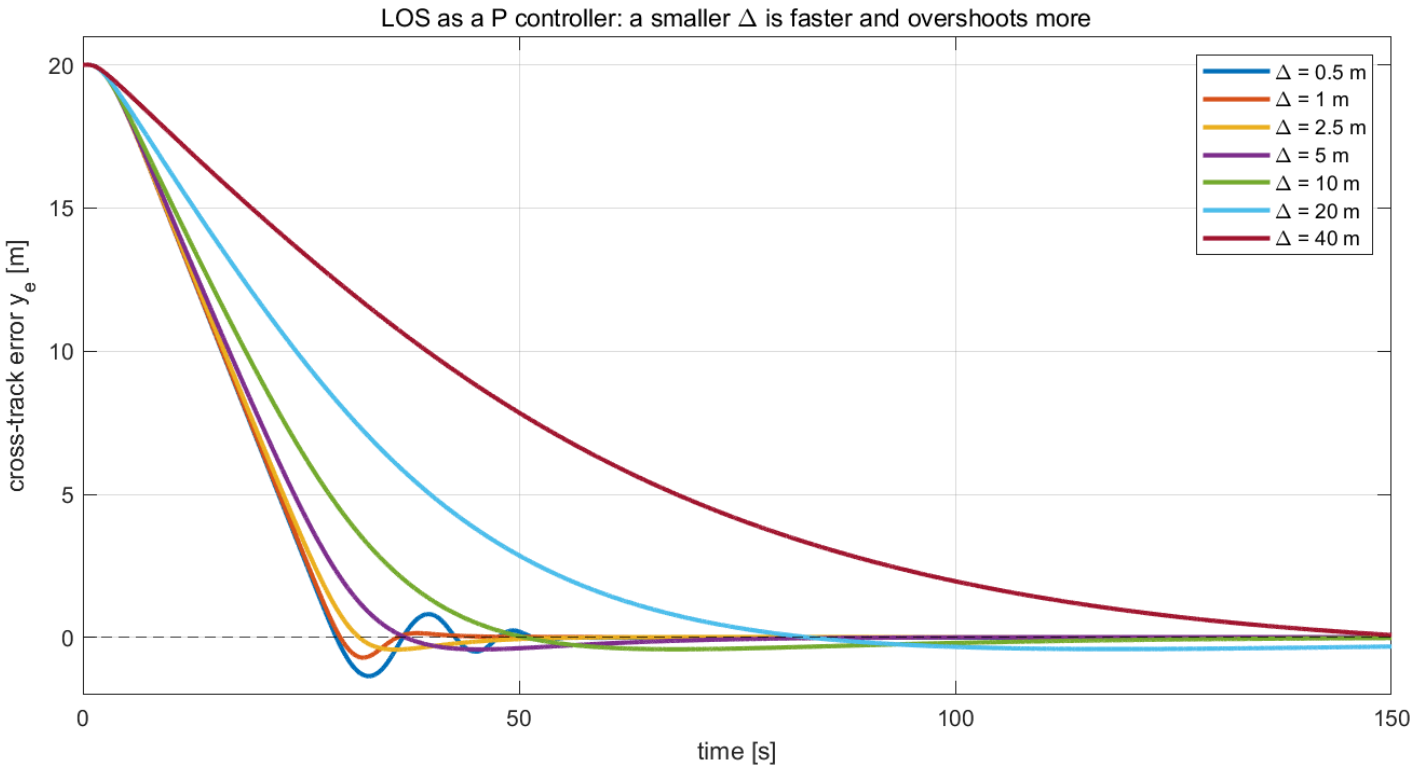
The model. `W05_D_LOS` — the guidance block switched to the LOS law. The waypoints are one 400 m leg, long enough for the transient to finish before any corner.

Opening and running.

```
W05_0_setup
WP_N = [0 400]'; WP_E = [0 0]';    % 긴 다리 하나 / one long leg
open_system('W05_D_LOS')           % Run — Delta = 5 m / the chosen value
Delta = 0.5;                       % Run — 지그재그로 다가간다 / it zigzags onto the path
Delta = 40;                        % Run — 한참 걸린다 / it creeps
W05_D_lookahead_distance           % 일곱 값을 한 번에 / all seven values at once
```

Expected output:

```
W05 Experiment 5-3  LOS, the look-ahead distance  (start 20 m east of the path)
  Delta [m]   P gain 1/Delta   within 1 m after [s]   overshoot [m]   zero crossings
  0.5                 2.000                   27.7            1.35                7
  1                   1.000                   27.9            0.70                2
  2.5                 0.400                   29.1            0.42                2
  5                   0.200                   32.4            0.41                2
  10                  0.100                   41.7            0.41                2
  20                  0.050                   65.2            0.40                1
  40                  0.025                  118.3            0.40                1
```



In the figure	Meaning
traces	the cross-track error against time from 20 m off the path, one per Δ

Reading the figure against the law.

Where to look	What is there	What it corresponds to
the **steepness** of each trace at the start	steeper the smaller Δ is	$K_p = 1/\Delta$: a larger gain pushes harder on the same error
the $\Delta = 0.5$ m trace **crossing zero** repeatedly	7 crossings, overshoot 1.35 m	too large a gain rings, as in Week 2 §2-6 — here the ringing is a zigzag across the path
the $\Delta = 40$ m trace	within 1 m only after 118.3 s	too small a gain is slow, and $1/40$ is a very small gain
the middle traces, $\Delta = 2.5$ to 10 m	all reach 1 m within 29 to 42 s with the same 0.41 m of overshoot	the flat part of the trade-off, where the choice is made
every trace, once on the path	it stays there	the law has no steady-state error to leave on a straight path with no current — §5-5 is where that changes

What the figure says

- The curves fan out exactly as a P gain sweep does: the small Δ drops fast and crosses back and forth; the large one creeps.

What to try	What to watch
`Delta = 0.2;` Run	the zigzag never stops: **44** crossings of the path and still 1.20 m off at the end of the leg. This is a K_p raised past its limit, with the hull's own turn rate as the slow element
`Delta = 100;` Run	one crossing and no zigzag at all, but the path is reached only after 286 s — a gain of 0.01 rad/m spends the whole leg converging

> ✎ **In class**
>
> - **Purpose** — recognise a familiar controller in an unfamiliar law.
> - **Point to** — the $\Delta = 0.5$ m trace crossing zero repeatedly; the $\Delta = 40$ m trace still 5 m off after 100 s.
> - **Ask** — "Which way does the gain go when Δ is halved?" Up: $K_p = 1/\Delta$ doubles.
> - **Take away** — tune Δ as K_p was tuned: the smallest value that does not swing.

5-4. When to move to the next leg

This section answers: at a corner, when does guidance switch to the next leg?

What is observed. The same mission is run with five switching distances, and the corners are where the tracks differ:

- with $R = 20$ m the vessel turns early and passes almost 20 m from the new leg,
- with $R = 1$ m it runs **past** the corner before turning, and sweeps 3.70 m outside the sharpest one,
- and the mission finishes sooner the larger R is: 305 s against 364 s.

What follows from it, one line at a time.

1. The switch is a test on the **along-track** coordinate: the guidance moves to the next leg when less than R of the active leg remains, $d - x_e < R$, with d the leg's length. This is the test MSS uses, and it is a distance along the path rather than a circle round the waypoint.

The two tests, and why they are not the same. Both are one line of code and they disagree:

$$\text{along-track:}\quad d - x_e < R \qquad\qquad \text{acceptance circle:}\quad \|\mathbf{p} - \mathbf{p}_{k+1}\| \le R$$

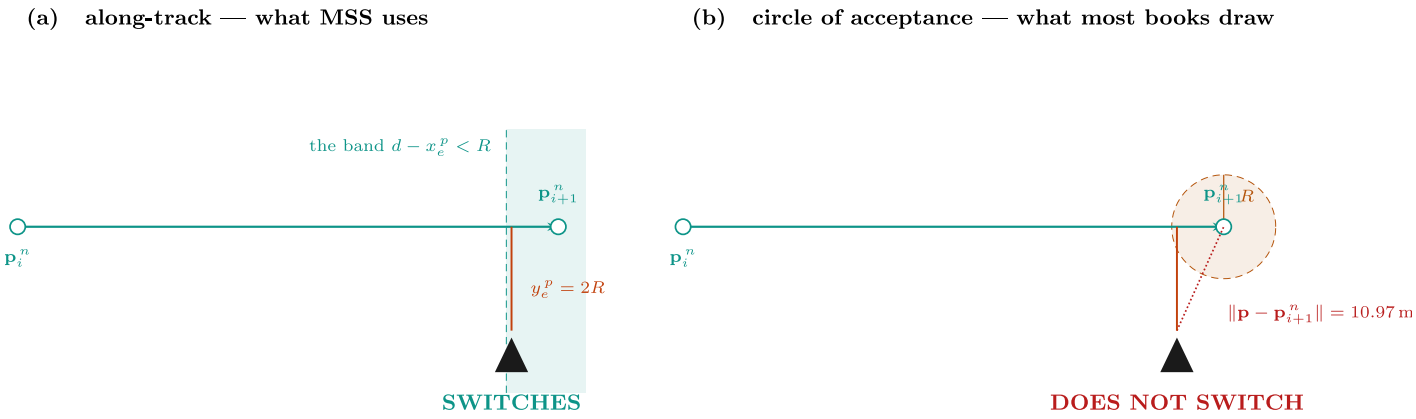
Symbol	Quantity	Value · source
d	the length of the active leg	m · $\|\mathbf{p}_{k+1} - \mathbf{p}_k\|$
x_e	how far along it the vessel has come	m · §5-1
$d - x_e$	what is **left to run** along the leg	m · negative once the vessel is past the waypoint
R	the switching distance	3 m · `R_switch`, chosen in line 4

Writing the true distance in path-frame components shows the relation exactly:

$$\|\mathbf{p} - \mathbf{p}_{k+1}\| = \sqrt{(d - x_e)^2 + y_e^2} \;\ge\; d - x_e$$

2. **On the path the two agree; off it they never do.** With $y_e = 0$ the square root collapses to $d - x_e$ and the tests are identical. With $y_e \neq 0$ the circle test is **always the stricter of the two**, by an amount that grows with how far off the path the vessel is — so a vessel that is running wide can satisfy the along-track test and fail the circle at the same instant.
3. **The circle can be missed altogether.** A vessel whose cross-track error never falls below R never enters the circle, passes the waypoint, and waits for an arrival that has already happened — for ever. The along-track test cannot fail that way: $d - x_e$ goes negative the moment the vessel crosses the line through the waypoint perpendicular to

the leg, whatever its cross-track error. This is not a hypothetical; Week 9 §9-5 loses a whole mission to it at 1.3 m/s of current.



The test looks only at how much of the leg is left, so being far off the path cannot defeat it: a vessel blown wide still moves on to the next leg.

```
if ( d - x_e < R_switch ) k = k + 1
```

The test is a true distance, so a vessel that never enters the circle waits for a waypoint it has already passed — and can wait for ever.

```
if ( norm(p - wp_next) < R ) k = k + 1
```

On the path the two agree exactly, because $y_e^p = 0$ makes $\|\mathbf{p}-\mathbf{p}_{i+1}^n\| = d - x_e^p$. Off it they do not: $\|\mathbf{p}-\mathbf{p}_{i+1}^n\| = \sqrt{(d-x_e^p)^2 + y_e^2} \geq d - x_e^p$, so **the circle test is always the stricter of the two**, by an amount that grows with y_e^p. Experiment 5-4 evaluates both on one position: the along-track remainder is 4.50 m and passes; the true distance is 10.97 m and fails. The one hard limit is $R < \min_k d_k$ — otherwise a leg is already "reached" before the vessel has travelled any of it.

Reading the figure.

Where to look	What is there	Which line predicts it
(a), the shaded band before the waypoint	the region where $d - x_e < R$	line 1: the test is a distance **along** the leg
(a), the vessel a whole $y_e = 2R$ off the path	still inside the band, and it switches	line 3: being off the path cannot defeat the along-track test
(b), the circle of radius R round the waypoint	the region where $\|\mathbf{p} - \mathbf{p}_{k+1}\| \leq R$	the other test, drawn
(b), the same vessel, outside the circle	10.97 m away, and it does **not** switch	line 2: the circle is the stricter test
the two code lines under the panels	`d - x_e < R_switch` against `norm(p - wp_next) < R`	one line of difference, and the whole behaviour
the caption's arithmetic	along-track remainder 4.50 m passes; true distance 10.97 m fails	line 2: $\sqrt{(d-x_e)^2 + y_e^2} \geq d - x_e$, here by 6.47 m

4. **There is one hard limit on R**, and it belongs to the along-track test: $R < \min_k d_k$. If R exceeds a leg's own length, then $d - x_e < R$ is true the instant the leg becomes active, and that leg is "reached" before the vessel has travelled any of it.
5. **A large R cuts the corner.** At the instant of the switch the vessel is still on the old leg, R before its end, so it is about R away from the new leg — and the new leg is what y_e is now measured against. Hence the largest distance is close to R itself in the last three rows.
6. **A small R carries the vessel past the corner.** The turn begins only at the corner, and the hull cannot turn instantly: at 0.77 m/s it sweeps outside, 3.70 m at the 135° corner for $R = 1$ m. The sharper the corner, the further it goes.
7. Lines 5 and 6 pull in opposite directions, so the best R is the one whose **worst** corner is smallest — $R = 3$ m, at 2.98 m.
8. Cutting corners is not wrong in itself; it also finishes sooner. The choice made here is to stay close to the path, and a survey that must cover its lines exactly would choose differently.

Measured by Experiment 5-4, below, on the five-waypoint mission with $\Delta = 5$ m:

R [m]	largest distance from the new leg at corners 1, 2, 3 [m]	last waypoint reached at [s]
1	2.06, 2.08, 3.70	363.5
3	2.98, 2.98, 2.29	352.8
5	4.98, 4.98, 3.53	344.0
10	9.97, 9.98, 7.04	328.0
20	19.97, 19.98, 14.08	305.4

Experiment 5-4 · Waypoint switching (10 min)

What it measures. Lines 5, 6 and 7: how far from the new leg each switching distance leaves the vessel at the three corners, and how long the mission then takes.

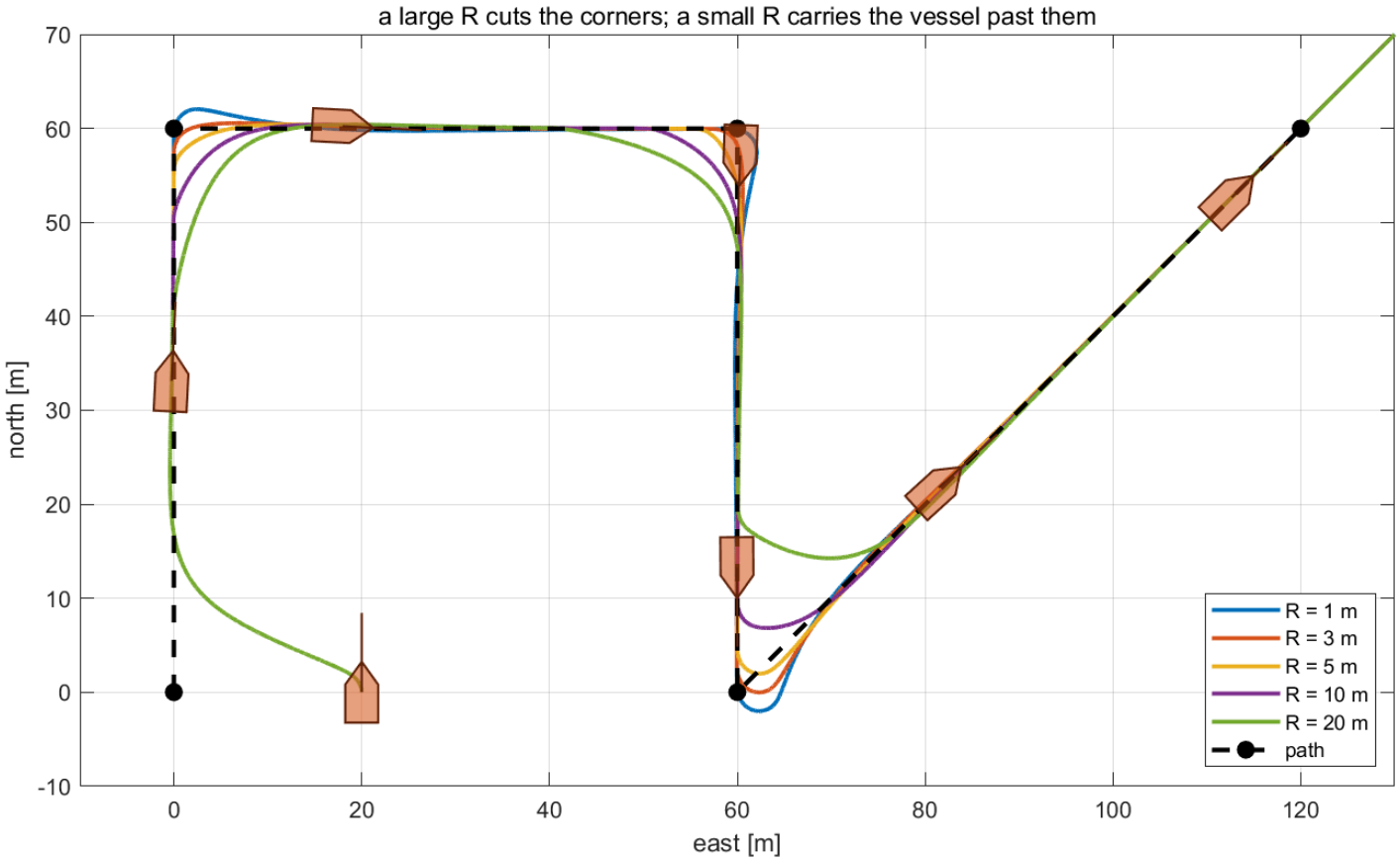
The model. `W05_E_switching` — the LOS law on the full five-waypoint mission, with Δ fixed at the value Experiment 5-3 chose. Only `R_switch` changes between runs, and the XY Graph shows the corners as they are rounded.

Opening and running.

```
W05_0_setup
open_system('W05_E_switching')     % Run — R = 3 m / the chosen value
R_switch = 20;                     % Run — 모퉁이를 크게 자른다 / it cuts the corners wide
R_switch = 1;                      % Run — 모퉁이를 지나쳐서 돈다 / it runs past them
W05_E_waypoint_switching           % 다섯 값을 한 번에 / all five values at once
```

Expected output:

```
W05 Experiment 5-4  waypoint switching on the mission  (LOS, Delta = 5 m, no current)
  R_switch [m]   largest distance from the new leg at corners 1, 2, 3 [m]   last waypoint at [s]
  1                      2.06   2.08   3.70                                  363.5
  3                      2.98   2.98   2.29                                  352.8
  5                      4.98   4.98   3.53                                  344.0
  10                     9.97   9.98   7.04                                  328.0
  20                    19.97  19.98  14.08                                  305.4
```



In the figure	Meaning
coloured tracks	the mission for $R = 1$ to 20 m; hulls drawn on $R = 3$ m
dashed line and dots	the path and its five waypoints

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
the $R = 20$ m track at the first corner	it leaves the leg some 20 m early	line 2: the switch happens R before the end of the leg
the largest-distance columns for $R = 5$, 10, 20	4.98, 9.97, 19.97 m — each equal to R	line 2 again, in numbers
the $R = 1$ m track at the **third** corner	the widest sweep of all, 3.70 m	line 3: the turn starts at the corner, and the 135° corner is the sharpest
the $R = 3$ m columns	2.98, 2.98, 2.29 m — the smallest worst case	line 4: the compromise between the two
the last column, top to bottom	$363.5 \rightarrow 305.4$ s	line 5: cutting corners shortens the mission
the straight parts between corners	every track lies on the path	§5-3: the switching distance decides the corners only

What the figure says

- The green track rounds every corner early and wide; the blue one runs past each corner before turning. $R = 3$ m sits between them.

What to try	What to watch
`R_switch = 50;` Run	almost as long as a 60 m leg: the guidance switches as soon as a leg begins, the vessel is never nearer than 49.98 m to the leg it is supposed to follow, and it reaches the last leg in 146.9 s by flying the diagonal
`Delta = 20;` `R_switch = 3;` Run	the worst corner grows to 7.93 m although R has not changed — a gain of $1/20$ cannot pull the vessel onto the new leg quickly, so Δ and R are read together

> ✎ **In class**
>
> - **Purpose** — the second tuning knob, set from what the corners look like.
> - **Point to** — the corner at (60, 0), where the tracks spread the most.
> - **Ask** — "Why does the largest distance equal R for large R?" At the switch the vessel is on the old leg, R before the corner — R from the new leg.
> - **Take away** — too early cuts, too late overshoots; choose from the worst corner.

5-5. A current, and the integral that removes its offset (ILOS)

This section answers: what does LOS do when a current pushes the vessel sideways, and how is it fixed?

What is observed. The same LOS law, the same Δ, with a current flowing east across a path running north:

- the vessel settles **parallel to the path but beside it** — 2.107 m off in a 0.3 m/s current, and it stays there,
- while pointing 22.8° into the current,
- and the offset grows with the current: 0.658, 1.344, 2.107, 3.012 m.

What follows from it, one line at a time.

1. To travel north across an eastward current the vessel must **point** west of north: the crab angle of Week 4 §4-4. Its course is along the path while its heading is not.
2. LOS produces that heading only through $\arctan(y_e/\Delta)$, so it can hold a heading away from π_p **only while** $y_e \neq 0$. This is Week 2 §2-6 line 7 once more: a proportional law makes its output out of error alone.
3. In the steady state the heading held equals the heading asked for, $\psi = \psi_d = \pi_p - \arctan(y_e/\Delta)$. Solving for the error,

$$y_e = \Delta \, \tan(\pi_p - \psi)$$

4. The offset is therefore **predictable, not mysterious**: it is Δ times the tangent of the crab angle the current forces. The measurements below match it to the millimetre (`verify_w05_guidance` check 3).
5. Reducing Δ would shrink the offset, by line 4 — but Δ was already chosen in §5-3 for the transient, and a smaller one zigzags. Two requirements, one knob: the standing answer of this course is to add the **integral** instead.

Measured by Experiment 5-5a, below, a current flowing east across a path north:

current [m/s]	y_e left [m]	heading held [deg]	$\Delta \tan(\text{heading})$ [m]
0.1	0.658	−7.5	0.658
0.2	1.344	−15.0	1.344
0.3	2.107	−22.8	2.107
0.4	3.012	−31.1	3.012

- The last two columns are line 4: prediction and measurement agree to the millimetre in every row. Heading is not course — the vessel points 22.8° into a 0.3 m/s current and travels due north.

Deriving the cure. Lines 1 to 5 say the offset exists and how large it is. They do not yet say what to replace the law with. That takes one more equation — the one this week has so far avoided writing, the **motion of the error itself**.

6. **How y_e moves.** Differentiate the definition of §5-1, $y_e = -(N - N_k)\sin\pi_p + (E - E_k)\cos\pi_p$, along a straight leg where π_p is constant. Writing the velocity in NED as $\dot N = U\cos\chi$ and $\dot E = U\sin\chi$, with U the speed over ground and χ the course,

$$\dot y_e = -U\cos\chi\,\sin\pi_p + U\sin\chi\,\cos\pi_p = U\,\sin(\chi - \pi_p)$$

Read it in words: **the cross-track error changes at the rate at which the vessel is travelling sideways relative to the path.** Nothing else in the vessel enters. And the course is the heading plus the crab angle, $\chi = \psi + \beta$, which is where the current comes in — Week 4 §4-4.

Symbol	Quantity	Value · source
U	speed over ground	≈ 0.77 m/s at $X_{ff} = 60$ N
χ	course, the direction actually travelled	$\chi = \psi + \beta$
β	crab angle, the sideways push of the current	22.8° at $V_c = 0.3$ m/s, Experiment 5-5a

> **★ Course and heading are two different angles, and the autopilot only knows one of them**
>
> The **course** χ is the direction the vessel is actually moving; the **heading** ψ is the direction its bow points. Their difference is the crab angle, and with a sway velocity v it is
>
> $$\chi = \psi + \beta_c, \qquad \beta_c = \arctan\frac{v}{u} = \arcsin\frac{v}{\sqrt{u^2+v^2}}$$
>
> A guidance law of the LOS family is naturally a **course** command: $\dot y_e = U\sin(\chi - \pi_p)$ contains χ and nothing else. What it is wired to decides what has to be done with that fact:
>
> | | **Course control** | **Heading control — this course** |
> |---|---|---|
> | the autopilot closes on | χ, measured as course over ground (GNSS) | ψ, measured by a compass or gyro (Week 4) |
> | the guidance output is | used directly: χ_d | converted: $\psi_d = \chi_d - \beta_c$ |
> | works at zero speed | no — χ is undefined when $U = 0$ | yes, which is why marine craft carry a compass |
> | the crab angle is | inside the loop, measured | **a disturbance in yaw**, outside it |
>
> The Otter of this course has the Week 4 heading autopilot, so it sits in the right-hand column: the guidance hands over ψ_d and whatever β_c is doing arrives as an unmeasured yaw disturbance. Experiment 5-5a measures it directly — the vessel holds -22.8° while making good due north, so $\beta_c = 22.8^\circ$ — and §5-5's whole problem is that the law has no way to be told that number.

7. **LOS, closed.** Substitute $\psi = \psi_d$ from §5-3 into line 6:

$$\dot y_e = U\,\sin\!\left(\beta - \arctan\frac{y_e}{\Delta}\right)$$

Set $\dot y_e = 0$ and the bracket must vanish, so $\arctan(y_e/\Delta) = \beta$ and $y_e = \Delta\tan\beta$. This is line 3 again, but arrived at from the **dynamics** rather than from a statement about steady state, and it says something stronger: that offset is **the equilibrium of the closed loop**, and the only one. It is not a transient that has not finished. Waiting does not remove it, and neither does any gain, because Δ is the only gain in the law and line 4 already priced it.

8. **What would have to be true instead.** For $y_e = 0$ to become an equilibrium, the bracket in line 7 must vanish *at* $y_e = 0$. Since $\beta \neq 0$, the arctan must be fed by something that is **not** y_e. Add a second term to it and give that

term a state of its own:

$$\psi_d = \pi_p - \arctan\!\left(\frac{y_e + \kappa\, y_{\text{int}}}{\Delta}\right) \qquad \Longrightarrow \qquad \dot{y}_e = U\,\sin\!\left(\beta - \arctan\frac{y_e + \kappa\, y_{\text{int}}}{\Delta}\right)$$

Now $y_e = 0$ is an equilibrium **if and only if**

$$\kappa\, y_{\text{int}}^{\star} = \Delta \tan\beta$$

and the right-hand side is, by line 3, **exactly the offset LOS was leaving**. That is the whole design, in one line: the integral does not cancel the current, and never learns what a current is. It converges to the number the old law was producing as an error, and produces it as a state instead. The measurement is the same number twice — $\mathbf{2.107}$ m of LOS offset, and $\kappa y_{\text{int}} \to \mathbf{2.10}$ m:

κ	$\kappa\, y_{\text{int}}$ at the end [m]	mean y_e, last 200 s [m]	peak $\lvert\dot{y}_{\text{int}}\rvert$
0.1	2.1013	0.0002	0.3949
0.3	2.1042	−0.0000	0.3361
1	2.1456	0.0015	0.8193
3	0.1187	0.1720	1.2337

Measured on one 900 m leg at $V_c = \mathbf{0.3}$ m/s, given 1100 s to settle, against $\Delta\tan\beta = \mathbf{2.1049}$ m for that run. The first three rows land on it to within 2 %; $\kappa = \mathbf{3}$ does not converge at all, which is line 11. The last column is measured here and read in line 9.

9. **How to drive the state there.** Line 8 fixes where y_{int} must end up but not how it gets there. Any update law will do provided it (a) is zero only when $y_e = 0$, so the equilibrium of line 8 is the one it stops at, and (b) does not run away while the vessel is far off the path, which is the windup of Week 2. Børhaug, Pavlov and Pettersen choose

$$\dot{y}_{\text{int}} = \frac{\Delta\, y_e}{\Delta^2 + (y_e + \kappa\, y_{\text{int}})^2}$$

Requirement (a) is immediate: the numerator carries the sign of y_e and vanishes only with it. Requirement (b) is the denominator's doing. While the integral is still small the argument is $\approx y_e$, and

$$\max_{y_e} \frac{\Delta\, y_e}{\Delta^2 + y_e^2} = \frac{1}{2} \quad \text{at} \quad y_e = \Delta$$

so however far off the path the vessel starts, the state cannot grow faster than half a unit per second — **anti-windup written into the law, not bolted onto it** as the back-calculation of Weeks 2 to 4 was. It is also why the update is fastest at $y_e = \Delta$ and slower both nearer and further: near the path there is little to integrate, and far from it the law is busy turning rather than correcting.

Symbol	Quantity	Value · source
y_{int}	the integral state	s; starts at 0
κ	integral constant — a **speed**, since κy_{int} is a length	$\mathbf{0.3}$ m/s, `W05_0_setup.m`
κ/Δ	the resulting I gain	$\mathbf{0.06}$ s^{-1}

10. **Small signals: it is a PI controller.** Near the path $\arctan x \approx x$ once more, so $\psi_d - \pi_p \approx -(1/\Delta)\, y_e - (\kappa/\Delta)\, y_{\text{int}}$ — proportional plus integral on the cross-track error, with $K_p = 1/\Delta$ from §5-3 and $K_i = \kappa/\Delta$. Everything Week 2 established about a PI controller applies here unchanged, including that the integral costs overshoot.

9b. **Why not the plain integral?** The obvious update is the one Week 2 used: integrate the error itself, $\dot{y}_{\text{int}} = y_e$, giving

$$\psi_d = \pi_p - \arctan\!\left(K_p\, y_e + K_i \int_0^t y_e(\tau)\,\mathrm{d}\tau\right), \qquad K_p = \frac{1}{\Delta}, \quad K_i = \frac{\kappa}{\Delta}$$

This is the **conventional ILOS** of the literature, and it satisfies requirement (a) — it stops only where $y_e = 0$. It fails requirement (b): far from the path it integrates at full rate. This mission starts 20 m off, so by the time the vessel reaches the path the state has been accumulating for half a minute, and it then has to unwind through the other side. Experiment 5-5c measures what that costs: at the same κ the plain integral does not follow the path at all, and tuned down until it does, it still swings 13.5 m where the nonlinear law swings 3.9 m. The denominator is the whole difference, and it is written into the law rather than bolted on.

10b. **Why not measure the crab angle instead?** If u and v are measured, β_c can be computed directly and subtracted, as the table above the derivation shows:

$$\psi_d = \pi_p - \arctan\frac{y_e}{\Delta} - \beta_c, \qquad \beta_c = \arcsin\frac{v}{\sqrt{u^2+v^2}}$$

It works, and it is the cleanest thing to do **when the measurement exists**. Three reasons the integral is used instead, and the third is the one that decides it here:

	Why it is a problem
the formula is singular at zero speed	β_c is undefined as $U \to 0$ and ill-conditioned just above it, exactly where a vessel manoeuvres near a berth
the measurement must exist	it needs v — a Doppler log, or course over ground differenced against the compass. The Otter of this course logs no v to the guidance block
it is never complete	roll and pitch coupling leaves a small offset even after β_c is subtracted, so an integral is usually added anyway

And the two are not really rivals: line 8 showed that the integral converges to $\kappa y_{\text{int}}^{\star} = \Delta \tan\beta$. **The integral reconstructs the number the measurement would have supplied**, from the path error alone, without a sensor and without a singularity at zero speed.

11. **And κ has a ceiling.** The bound in line 9 holds while y_{int} is small; once κy_{int} is comparable to y_e and opposite in sign the denominator shrinks again and the update is no longer capped. Measured peak $|\dot{y}_{\text{int}}|$: 0.34 at $\kappa = 0.3$, but 0.82 at $\kappa = 1$ and 1.23 at $\kappa = 3$ — past the bound. The last of those never settles, as the table in line 8 shows. The stability proof in the source paper carries a condition on κ for exactly this reason; this course chooses κ by measurement instead, in §5-6.

Sources. The law and its Lyapunov analysis: Børhaug, Pavlov and Pettersen, "Integral LOS control for path following of underactuated marine surface vessels in the presence of constant ocean currents", *47th IEEE Conference on Decision and Control*, 2008, pp. 4984–4991, DOI 10.1109/CDC.2008.4739352 — they prove UGAS/ULES for a 3-DOF ship model in cascade with the guidance law. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., 2021, §12.3 gives the same law with the notation used here and the proof on p. 362; Lekkas and Fossen (2014) is where the form with guaranteed stability properties is set out. The MSS implementations are `ILOSpsi.m` and `ILOSchi.m`.

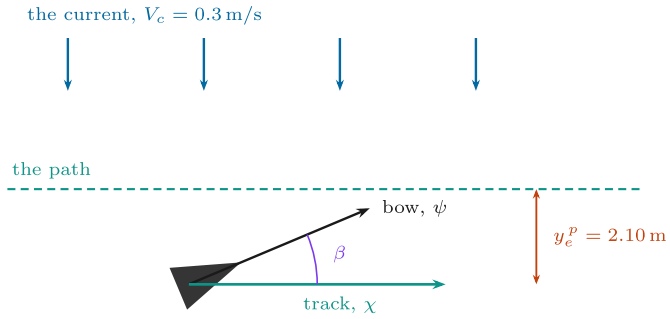
> **The MSS integrator is the same idea with a different scaling**
>
> `ILOSchi.m` updates the state as $\dot{y}_{\text{int}} = U\, y_e / \sqrt{\Delta^2 + (y_e + \kappa y_{\text{int}})^2}$ — the speed over ground in the numerator and a square root in the denominator, where the form used here and in Fossen's lecture notes has Δ and no root. Both vanish only with y_e and both shrink as the error grows, which is all lines 8 and 9 require; the difference is the units of κ and how the rate scales with speed. This course keeps the book's form so that κ can be read as a speed and compared with the current directly.

Both laws end up pointing the bow upstream by the same angle. They differ in what is left over.

Same vessel, same path, same current: $V_c = 0.3\,\text{m/s}$ on the beam, so the vessel must hold $\beta = 22.8^\circ$ of crab and $\Delta \tan\beta = 2.10\,\text{m}$ with $\Delta = 5\,\text{m}$. Experiment 5-5a measures the offset at 2.107 m. **The bow tilt is the same in both panels — only one of them is on the path.**

1 LOS

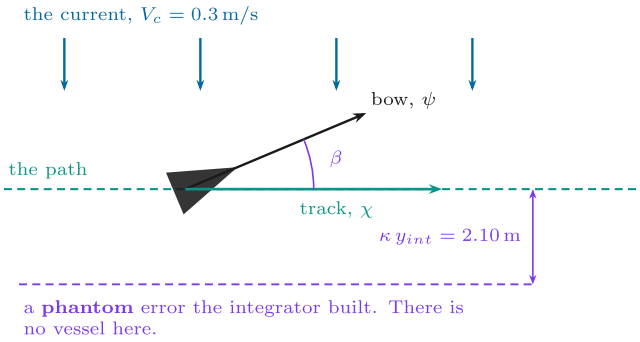
the lean is spent on the drift



The bow is tilted upstream by 22.8° — but that tilt *is* the $\arctan(y_e^p/\Delta)$ term, and it is entirely used up cancelling the drift. Nothing is left to close y_e^p. The vessel runs **parallel** to the path, 2.107 m to the side of it, for ever (Experiment 5-5a).

2 ILOS

an error that is not there, so the real one can be zero



The law reads $y_e^p + \kappa y_{int}$, not y_e^p. The integrator fills until its phantom error equals the old 2.10 m offset, and only then can the **real** error be zero — measured at 0.011 m with $\kappa = 0.3$ (Experiment 5-5b). It never learns what a current is; it only knows the error must vanish.

The one idea to take away. A current cannot be removed by steering harder. It must be **answered** by pointing the bow upstream. LOS points upstream but spends the whole lean on the drift, so an offset is left over; ILOS buys that lean from a second term and hands the arctan its original job back.

And the anti-windup is in the law, not bolted on. The integrator's own update is $\dot{y}_{int} = \Delta y_e^p / (\Delta^2 + (y_e^p + \kappa y_{int})^2)$, whose denominator grows with the error — so the state barely fills while the vessel is far off the path, and fills fastest just where it should, close to it. Weeks 2 to 4 had to add back-calculation for the same effect.

Sources. LOS — Fossen (2021), *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., §12.3. ILOS — Børhaug, Pavlov and Pettersen (2008), *Proc. 47th IEEE CDC*, pp. 4984–4991. The numbers in the panels are §5-5's measurements, not illustrations.

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
both panels, the bow arrow	tilted upstream by the **same 22.8°**	line 1: the crab angle is set by the current and the speed, not by the law
panel 1, where the hull sits	below the path, and its track is **parallel** to it	line 2: LOS can hold that tilt only while $y_e \neq 0$
panel 1, the red marker	$y_e = 2.10$ m, which is $\Delta \tan\beta$	line 3: the offset is the tangent, measured at **2.107** m
panel 2, where the hull sits	**on** the path, with the same tilt	line 8: the integral now supplies the angle, so the error need not
panel 2, the dashed line below the path	$\kappa y_{\text{int}} = 2.10$ m — a **phantom** error, with no vessel on it	line 8: the law reads $y_e + \kappa y_{\text{int}}$, and that sum is what must vanish
the two together	nothing about the current appears in either law	line 8: ILOS never learns what a current is; it converges to the number the old law left as an error

What the figure says

- The integral does not fight the current. It **replaces the error the current was producing** with one of its own, so that the arctan keeps leaning the bow by exactly as much as before while the real error goes to zero.

Measured by Experiment 5-5b at 0.3 m/s, with $\Delta = 5$ m:

κ	mean y_e, last 100 s [m]	overshoot [m]
LOS ($\kappa = 0$)	2.107	0
0.1	0.009	0.03
0.3	0.011	1.79
1	−0.011	4.10
3	0.169	6.32

- Every κ removes the offset — line 8 — and a larger one overshoots more, which is the price of every integral since Week 2. On a single long leg $\kappa = 0.1$ is the cleanest; §5-6 chooses on the mission, where the legs are short and the integral has less time.

Experiment 5-5a · The offset a current leaves under LOS (7 min)

What it measures. Lines 3 and 4: the error LOS settles at in four currents, against $\Delta \tan$ of the heading it holds.

The model. `W05_F_LOS` — the LOS law of §5-3 on one straight 400 m leg, long enough for the offset to settle. Only the current changes between runs.

Opening and running.

```
W05_0_setup
WP_N = [0 400]'; WP_E = [0 0]'; V_c = 0.3;     % 동쪽으로 0.3 m/s / 0.3 m/s to the east
open_system('W05_F_LOS')           % Run — 경로와 나란히, 2.1 m 옆으로 / parallel, 2.1 m beside it
V_c = 0.4;                         % Run — 더 비스듬히 서고 더 벌어진다 / further off
W05_F_LOS_in_a_current             % 조류 네 가지를 한 번에 / all four currents at once
```

Expected output:

```
W05 Experiment 5-5a  LOS in a current flowing east  (Delta = 5 m)
  V_c [m/s]   error left [m]   heading held [deg]   Delta*tan(heading) [m]
  0.1                 0.658                -7.5                    0.658
  0.2                 1.344               -15.0                    1.344
  0.3                 2.107               -22.8                    2.107
  0.4                 3.012               -31.1                    3.012
```

- The last two columns are line 4, to the millimetre in every row. The vessel points 22.8° into a 0.3 m/s current and travels due north — Week 4 §4-4, now costing a path error.

Experiment 5-5b · The integral that removes it (8 min)

What it measures. Lines 8 to 11: the same current with the integral switched on, for four values of κ — what each one costs in overshoot, whether κy_{int} really converges to the offset LOS was leaving, and where the update-rate bound of line 9 stops holding.

The model. `W05_F_ILOS` — the same guidance block with the integral state of the law above. The black line in the figure is the run of Experiment 5-5a, for comparison.

★ What to open, and what to read inside it

The integral of this week is **not** a Simulink `Integrator` block. It lives inside the `guidance` MATLAB Function, because its update depends on the very quantity it is producing — and that is easier to read as two lines than as a loop of blocks.

Double-click `guidance` in `W05_F_ILOS` and look for three things:

```
persistent k y_int                    % 이 블록이 기억하는 것 둘: 지금 다리, 적분 상태
if isempty(k), k = 1; y_int = 0; end  % Run 을 누를 때마다 처음으로 돌아간다

psi_d = pi_p - atan(y_e/Delta + (kappa/Delta)*y_int);          % ① 법칙
y_int = y_int + h * Delta*y_e / (Delta^2 + (y_e + kappa*y_int)^2);   % ② 적분
```

Read this	And see
`persistent k y_int`	the block has **memory**: the active leg and the integral state. Nothing else in the guidance block does
line ① against §5-3's law	one extra term inside the same arctan — that is the whole of ILOS
line ②'s denominator	$\Delta^2 + (y_e + \kappa y_{\text{int}})^2$, growing with the error: the anti-windup of line 9, with no saturation block and no back-calculation gain
`kappa = 0`	line ② still runs but contributes nothing to line ①, so the block becomes exactly the LOS of §5-3. Not approximately: run this way it reproduces `W05_F_LOS` to 0.000×10^{0} m, which is how Experiment 5-5a and this one are kept comparable

The block is discrete at h (`set_mlfcn` 's sample-time argument), because a block with memory must be told how often to update it.

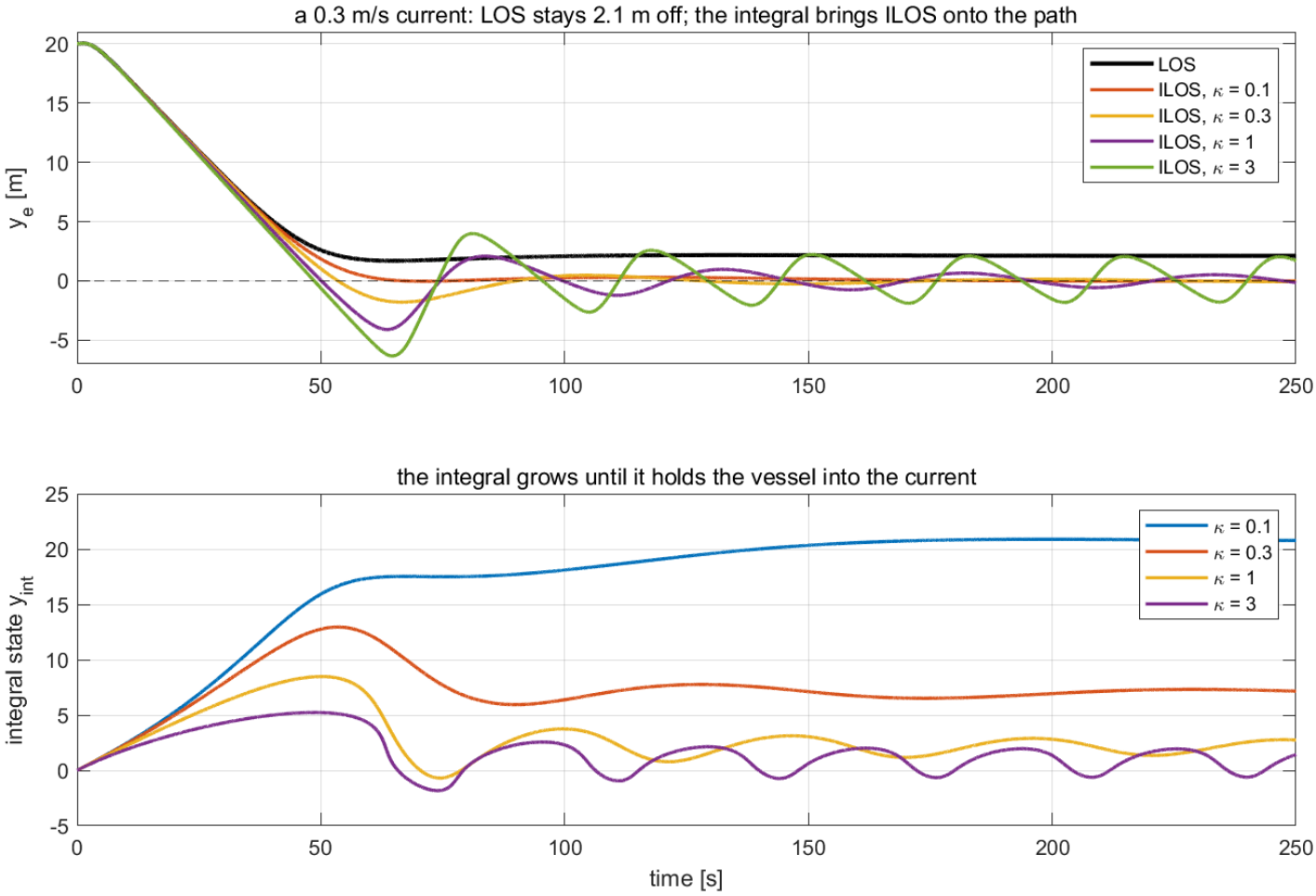
Opening and running.

```
W05_0_setup
WP_N = [0 400]'; WP_E = [0 0]'; V_c = 0.3;
open_system('W05_F_ILOS')          % Run — 같은 조류, 오차가 사라진다 / the same current, no offset
open_system('W05_F_ILOS/guidance') % 적분 두 줄을 직접 본다 / the two lines above
kappa = 1;                         % Run — 더 빨리 없애지만 더 넘어간다 / faster, and past it
kappa = 0;                         % Run — 같은 블록이 LOS 가 된다 / the same block becomes LOS
kappa = 0.3;
W05_F_ILOS_removes_it              % kappa 네 가지를 한 번에 / all four values at once
```

Expected output:

```
W05 Experiment 5-5b  ILOS at 0.3 m/s  (I gain = kappa / Delta)
  kappa   mean y_e, last 100 s [m]   overshoot [m]   integral state at 400 s
  0.1                       0.009            0.03                    20.95
  0.3                       0.011            1.79                     7.03
  1                        -0.011            4.10                     2.35
  3                         0.169            6.32                    -0.50

  line 8   the LOS offset is Delta*tan(beta) = 2.1049 m;  does kappa*y_int reach it?
    kappa   kappa*y_int at the end [m]   mean y_e, last 200 s [m]   max |dy_int/dt|
    0.1                    2.1013                   0.0002            0.3949
    0.3                    2.1042                  -0.0000            0.3361
    1                      2.1456                   0.0015            0.8193
    3                      0.1187                   0.1720            1.2337
```



In the figure	Meaning
top	the cross-track error: LOS in black, ILOS for four values of κ
bottom	the integral state of each ILOS run

Reading the figure against the derivation.

Where to look	What is there	Which line predicts it
top panel, the black LOS line	flat at 2.107 m, for ever	lines 2 and 3: the error is what produces the lean, so it cannot go
the LOS table, last two columns	equal in all four rows	line 4: $y_e = \Delta \tan$(crab angle)
top panel, every coloured trace	reaches zero and stays	line 8: the integral supplies the lean instead
bottom panel, the states **levelling off** at 20.95, 7.03, 2.35	different values, same effect	line 8: what is fixed is the product $\kappa\, y_{\text{int}}$, so a smaller κ needs a proportionally larger state — and $0.1 \times 20.95 = 0.3 \times 7.03$ to two decimals
the second table, second column	2.1013, 2.1042, 2.1456 against the LOS offset 2.1049	line 8, the point of the whole section: **the integral converges to the number LOS was leaving as an error**
the same table, $\kappa = 3$	0.1187, and y_e still at 0.17 m after 1100 s	line 11: this κ does not converge, so the row is not a measurement of an equilibrium
top panel, the overshoot at $\kappa = 1$ and 3	4.10 m and 6.32 m	the price of every integral since Week 2 §2-8: late correction arrives after it was needed
the first seconds of every trace, from 20 m off	no wind-up, no spike	line 9: while y_e is large the denominator is large and the state hardly moves
the last column at $\kappa = 0.1$ and 0.3	0.3949 and 0.3361, both under $\frac{1}{2}$	line 9: the bound holds while the state is small
the last column at $\kappa = 1$ and 3	0.8193 and 1.2337, both past $\frac{1}{2}$	line 11: once κy_{int} is large and opposed to y_e the denominator shrinks again and the bound is gone

What the figure says

- LOS settles 2.1 m off the path; every ILOS settles on it, and a larger κ gets there by swinging past.

What to try	What to watch
`V_c = 0.5;` on `W05_F_LOS`	4.234 m off, holding -40.3° — and $5 \tan 40.3^\circ = 4.234$ m: line 4 holds at any current
`Delta = 2.5; V_c = 0.3;` on `W05_F_LOS`	half the look-ahead distance halves the offset, 1.054 m at the same -22.9° heading, exactly as line 4 says. It also sharpens the transient, which is why §5-6 fixes Δ on the transient and leaves the current to κ

> ✎ **In class**
>
> - **Purpose** — the same story as the P and I of Week 2, now on a path.
> - **Point to** — the black line holding 2.1 m; the integral states levelling off at different values that all produce the same steady lean into the current.
> - **Ask** — "Why does LOS not reach the path, when its heading is exactly what it commanded?" To hold the line across the current, the vessel must point into it, and LOS only commands that while $y_e \neq 0$.
> - **Take away** — a steady push leaves an error under P; the integral removes it.

Experiment 5-5c · The plain integral against the nonlinear one (8 min)

What it measures. Line 9b: the same aiming law, the same current, the same mission, with the integral state updated two different ways. The aiming expression is identical in the two runs — $\psi_d = \pi_p - \arctan\left[(y_e + \kappa y_{\text{int}})/\Delta\right]$ — so

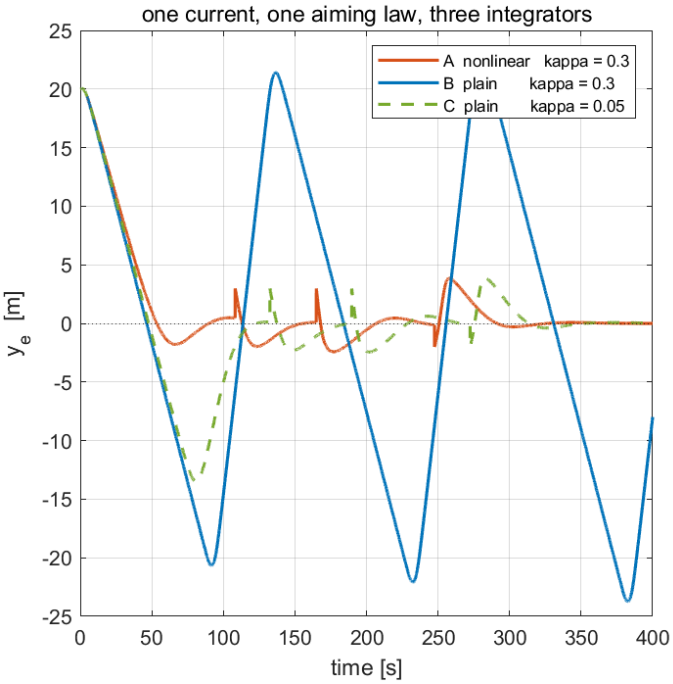
every difference between them belongs to the integrator.

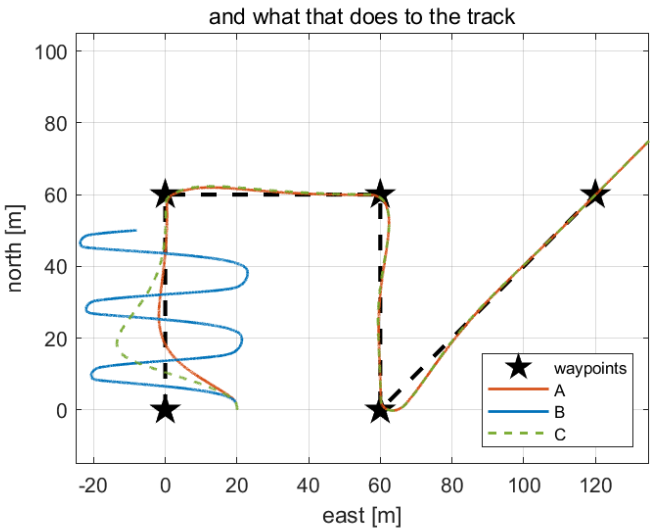
The model. `W05_F_ILOS` with `law = 4` for the plain integral and `law = 3` for the nonlinear one. Both laws live in the same `guidance` block, one above the other, and the flag chooses (§5-1b).

```
W05_0_setup
W05_F2_two_integrators
```

Actual output:

```
  W05 Experiment 5-5c  the two integrators  (current 0.3 m/s towards east, Delta = 5 m)
    run                          worst swing after the first crossing [m]   |y_e| mean over the last 100
s [m]
    A  nonlinear   kappa = 0.3                          3.85                                  0.071
    B  plain       kappa = 0.3                         23.73                                 11.529
    C  plain       kappa = 0.05                        13.48                                  0.203
```





In the figure	Meaning
left, the orange line (A)	the nonlinear update at $\kappa = 0.3$: across the path once, then within a metre of it
left, the blue line (B)	the plain integral **at the same** κ: it never follows the path, it circles it
left, the green dashed line (C)	the plain integral tuned down to $\kappa = 0.05$, which is the best this law does here
right	the same three as tracks: B weaves across the first leg, C overshoots into it, A joins it

Reading the figure against the derivation.

Where to look	What is there	Which line it settles
run B, the first 60 s	the error falls from 20 m to zero as fast as A does	line 9b: while y_e is large the two integrators are not what is steering — the arctan is
run B, after the first crossing	a swing of 23.7 m to the other side, then another, and another	line 9b: the state wound up during the approach and the unwinding carries the vessel through
run B, the last 100 s	mean $\lvert y_e \rvert = 11.5$ m — it never settles	at this κ the plain law is not a path follower at all
run C against run B	the same law, κ six times smaller, and it settles at 0.20 m	the plain integral can be made to work, by giving up gain
run C against run A	13.5 m of swing against 3.9 m, with A's steady error **three times smaller**	line 9: the denominator is what lets κ be large enough to reject the current quickly without winding up on the way in
all three, the last 100 s	0.071, 11.529, 0.203 m	line 8 holds for every integrator that stops only at $y_e = 0$; what differs is whether it gets there

What the figure says

- Both laws are "LOS plus an integral", and they differ in one line. That line decides whether κ may be chosen for the current or must be chosen for the approach.

What to try	What to watch
`E0 = 2;` then rerun	the start is now 2 m off instead of 20, the plain integral has nothing to wind up on, and B and A become nearly indistinguishable — which is the clearest statement of what the denominator is for
`kappa = 0.05` with `law = 3`	the nonlinear law with C's gain: it loses the current more slowly and keeps the clean transient

> ✎ **In class**
>
> - **Purpose** — show that an anti-windup written into a law is not a detail of implementation.
> - **Point to** — the blue trace crossing zero at full speed every time.
> - **Ask** — "Both integrators stop only at $y_e = 0$. Why do they not end up in the same place?" Because the equilibrium of line 8 says nothing about the path taken to it, and the plain one arrives with a state that has to be unwound.
> - **Take away** — the same equilibrium, two transients; the one that is safe to tune is the one with the denominator.

5-6. The tuning order, applied to guidance

This section answers: in what order are the three parameters chosen?

The order of Week 2 §2-12: the proportional part first, the integral only for an error that remains.

Step	What is chosen	From what measurement	Choice
1	Δ, the P gain	Experiment 5-3: fast without swinging on a straight leg	5 m
2	R, the switching distance	Experiment 5-4: the smallest worst corner	3 m
3	κ, the I gain	Experiment 5-6: the mission in a 0.3 m/s current	below

Measured by Experiment 5-6, below, mean $\lvert y_e \rvert$ on each leg from 25 s after entering it:

κ	leg 1	leg 2	leg 3	leg 4	sum of legs 2–4 [m]	last waypoint at [s]
0 (LOS)	3.29	0.12	2.14	1.38	3.64	330.4
0.1	2.04	1.22	1.33	0.55	3.10	337.0
0.3	2.29	0.40	0.38	0.14	**0.92**	338.8
0.5	2.44	0.30	0.48	0.15	0.93	342.6
1	2.62	0.24	0.66	0.18	1.07	349.8

- Leg 1 includes the 20 m start offset and is left out of the comparison. LOS leaves 2.14 m and 1.38 m on the legs that cross the current and almost nothing on leg 2, which runs with it.
- On 60 m legs $\kappa = 0.1$ is too slow. $\kappa = 0.3$ and 0.5 are nearly equal; Experiment 5-5b showed a larger κ overshoots more, so $\kappa = 0.3$.

The result, $\Delta = 5$ m, $R = 3$ m, $\kappa = 0.3$, is the default in `W05_0_setup.m`. No model of the vessel was used; only what the Scope and the track showed.

> **What is left out this week**
>
> The previous version of this week derived the laws in full, proved their stability with Lyapunov functions, and added a third law, ALOS, which estimates the crab angle directly. That material is kept in `W05_simulink/_previous_version/` for reference; nothing in this week's tuning depends on it.

Experiment 5-6 · The tuning order (15 min)

What it measures. Step 3 of the order above, on the mission the guidance is actually for: mean $|y_e|$ leg by leg, in a current, for five values of κ — and the time the mission takes.

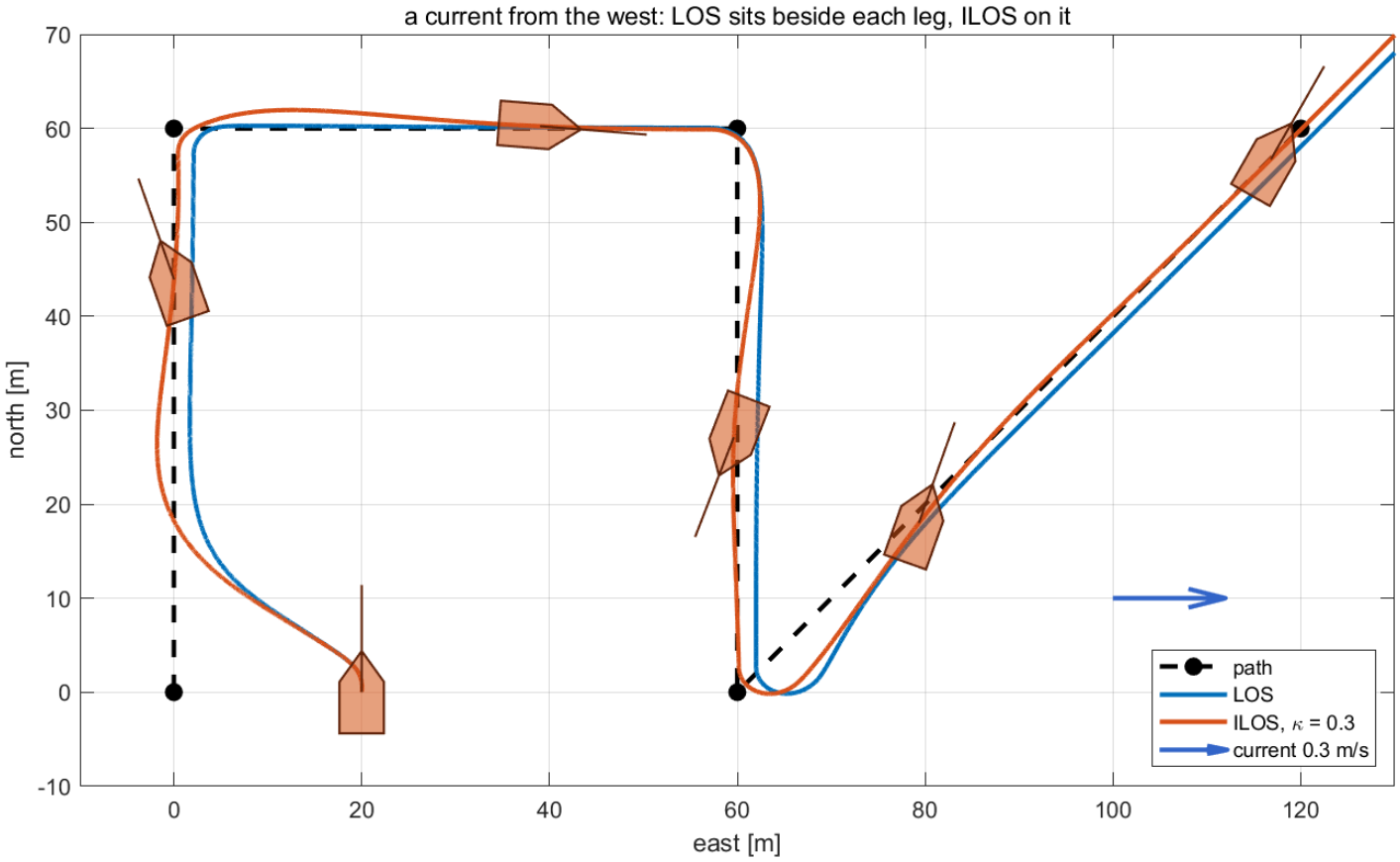
The model. `W05_G_tuning` — ILOS on the five-waypoint mission with Δ and R already fixed by Experiments 5-3 and 5-4. Only κ changes.

Opening and running.

```
W05_0_setup
V_c = 0.3; T_final = 450;          % 동쪽 조류 / a current to the east
open_system('W05_G_tuning')        % Run — kappa = 0.3 / the chosen value
kappa = 0;                         % Run — LOS 로 돌아간다 / back to LOS: it runs beside the path
W05_G_tuning_by_hand               % 다섯 값을 다리별로 / all five values, leg by leg
```

Expected output:

```
W05 Experiment 5-6  the tuning order  (Delta = 5 m, R_switch = 3 m, current 0.3 m/s east)
  kappa   mean |y_e| on legs 1, 2, 3, 4 [m]   sum of legs 2-4 [m]   last waypoint at [s]
  0           3.29   0.12   2.14   1.38                 3.64                 330.4
  0.1         2.04   1.22   1.33   0.55                 3.10                 337.0
  0.3         2.29   0.40   0.38   0.14                 0.92                 338.8
  0.5         2.44   0.30   0.48   0.15                 0.93                 342.6
  1           2.62   0.24   0.66   0.18                 1.07                 349.8
```



In the figure	Meaning
blue	LOS: beside each leg that crosses the current
orange with hulls	ILOS, $\kappa = 0.3$: on the legs; the hulls point into the current while the track stays on the path
arrow	the current, 0.3 m/s towards the east

Reading the figure and the table against the three steps.

Where to look	What is there	Which step or section it settles
the blue track beside legs 3 and 4	2.14 m and 1.38 m of mean error	§5-5 lines 2 to 4: the offset LOS must leave, now on a mission
the blue track **on** leg 2	0.12 m only	that leg runs with the current, so there is almost no crab angle to hold
the orange track on every leg	0.40, 0.38, 0.14 m	§5-5 line 7: the integral holds the lean instead
the hulls on the orange track	pointing into the current while the track runs along the leg	Week 4 §4-4: heading is not course, seen directly
the $\kappa = 0.1$ row	3.10 m, barely better than LOS	short legs: the value that was cleanest on 400 m has no time to act on 60 m
the last column, $330 \rightarrow 350$ s	the mission grows slower as κ grows	the integral's swing costs distance as well as accuracy
leg 1 in every row	2.0 to 3.3 m, and left out of the sum	it contains the 20 m start offset, which is a transient and not a following error

What the figure says

- The tuned ILOS follows the legs; the hulls show the crab angle it holds to do so.

What to try	What to watch
`kappa = 3;` Run	the swing of Experiment 5-5 now happens at every corner: mean $\lvert y_e \rvert$ after 60 s is 1.18 m against 0.64 m at $\kappa = 0.3$, and the last waypoint is reached at 385.9 s instead of 338.8 — **both slower and less accurate**
`V_c = 0; kappa = 0.3;` Run	with no current there is nothing to find: 0.30 m against 0.20 m for LOS, and the same mission time. The integral is insurance, and it is not free

> ✎ **In class**
>
> - **Purpose** — close the week with the full order on a mission, and see Week 4 §4-4 in the picture.
> - **Point to** — the hull on the southbound leg, pointing west of south while the track runs due south.
> - **Ask** — "Why is κ chosen on the mission and not on the long leg of Experiment 5-5b?" Short legs leave little time to integrate; the gain that is best on 400 m is too slow on 60 m.
> - **Take away** — choose each parameter on the task it will face, in the order P, switching, I.

5-7. Both loops at once: the speed of Week 3 and the path of Week 5

What is observed. Every model up to this point has pushed the hull with a constant surge force, `X_ff = 60` N. Nothing in the week ever asked what speed that produces. The answer is 0.767 m/s, and it is not a chosen number at all: it is the speed at which the drag of the Otter happens to balance 60 N. Ask for a different speed and there is no way to ask. Put the Week 3 speed loop in front of the same allocator and the vessel holds 1.000 m/s instead, and reaches the last leg of the mission at 189.7 s rather than 244.9 s.

The result. The two controllers do not meet, and never exchange a signal. They meet in the **allocator**, because there are two propellers and the surge force and the yaw moment must both come out of the same two. Writing that sharing down gives one 2×2 system, its inverse, and a limit on the yaw moment that **depends on the surge force**:

$$X = T_L + T_R, \qquad N = y_p\,(T_L - T_R) \qquad \Longrightarrow \qquad T_L = \frac{X}{2} + \frac{N}{2y_p}, \qquad T_R = \frac{X}{2} - \frac{N}{2y_p}$$

$$\lvert N \rvert \;\le\; N_{\lim}(X) \;=\; \min\left\{\, 2y_p\left(k_{pos}\,n_{\max}^2 - \tfrac{X}{2}\right),\;\; 2y_p\left(\tfrac{X}{2} + k_{neg}\,n_{\min}^2\right) \right\}$$

Symbol	Quantity	Value · source
X	surge force asked for	60 N constant, or the speed loop's output; `W05_0_setup`
N	yaw moment asked for	the autopilot's output, §5-1
T_L, T_R	thrust of the port and starboard propeller	N
y_p	half the distance between the two pontoons	0.3950 m, `otter.m`
k_{pos}, k_{neg}	thrust coefficients ahead and astern	0.011080, 0.006445; `otter.m`
$n_{\max}, n_{\min}$	shaft-speed limits	103.931, -101.737 rad/s; `otter.m`
$N_{\lim}(60)$	the moment left at the old constant force	70.85 N·m, Experiment 5-7
$N_{\lim}(120)$	the moment left at full surge force	47.15 N·m, Experiment 5-7

Derivation.

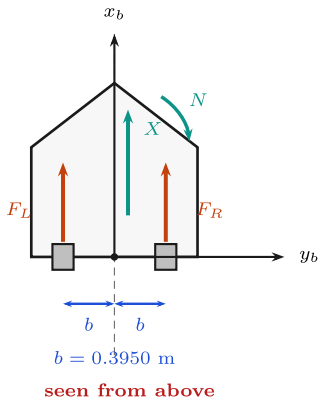
1. **Where the two forces come from.** The Otter has two fixed propellers, one on each pontoon, both pointing forward. Each produces a thrust T_i along the body x axis. Nothing steers; the hull turns because one side pushes harder than the other.

2. **The two forces they can make.** Body y is positive to starboard, so the port propeller sits at $y = -y_p$ and the starboard one at $y = +y_p$. A force T along x applied at y makes a yaw moment $-yT$. Summing both:

$$\underbrace{\begin{bmatrix} X \\ Y \\ N \end{bmatrix}}_{\tau} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 0 \\ y_p & -y_p \end{bmatrix}}_{\mathbf{B}} \begin{bmatrix} T_L \\ T_R \end{bmatrix}$$

The middle row is zero, and that row is the whole story of the week: **this hull cannot make a sideways force.** It is why guidance must turn the vessel to move it sideways, and why §5-5's current is fought with a heading and not with a thrust. `_tools/otter_B.m` builds this $\mathbf{B}$ from the column rule and returns exactly the matrix above.

This is the same matrix Week 3 §3-6 read off the hull and Week 4 §4-7 used with a live moment, written here with y_p in place of b. The figure is the one those weeks used, and it is worth re-reading now that **both** of its inputs are outputs of controllers:



1. read the three forces off the picture

$$\begin{bmatrix} X \\ Y \\ N \end{bmatrix} = \begin{bmatrix} F_L + F_R \\ 0 \\ b\,(F_L - F_R) \end{bmatrix}$$

both push forward, so they **add**

neither pushes sideways: **this row is empty, always**

opposite arms, so they **subtract**

2. drop the empty row — it is square now

$$\begin{bmatrix} X \\ N \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ b & -b \end{bmatrix}}_{\mathbf{T}} \begin{bmatrix} F_L \\ F_R \end{bmatrix}$$

$\mathbf{T}$ is **geometry**, nothing else: it never changes while the propellers stay bolted where they are

3. invert it — that is the allocator

$$\begin{bmatrix} F_L \\ F_R \end{bmatrix} = \mathbf{T}^{-1} \begin{bmatrix} X \\ N \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2b} \\ \frac{1}{2} & -\frac{1}{2b} \end{bmatrix} \begin{bmatrix} X \\ N \end{bmatrix}$$

$\mathbf{T}^{-1}$: known

$\begin{bmatrix} \frac{1}{2} & \frac{1}{2b} \\ \frac{1}{2} & -\frac{1}{2b} \end{bmatrix}$: what the two controllers asked for

$\det \mathbf{T} = -2b \neq 0$, **so the inverse exists and the split is unique: each propeller takes half of X, and the moment is added to one and taken from the other**

In the figure	What it means in this section
the two red arrows	both propellers push forward, which is the row $\begin{bmatrix} 1 & 1 \end{bmatrix}$
the two arms marked b	here called y_p: equal, opposite, and the reason the third row subtracts
the teal N arrow	**clockwise seen from above**, because z_b points down and $N > 0$ swings the bow to starboard
the empty middle row	why §5-5's current is fought with a heading and never with a sideways thrust
line 3, $\mathbf{T}^{-1}$	the allocator. Until this week one of its two inputs was a constant; from §5-7 both come from controllers

- Written out at $y_p = 0.395$ m as $\begin{bmatrix} 1 & 1 \\ 0.395 & -0.395 \end{bmatrix}$, this matrix sits as a Constant block inside the `Allocation` subsystem of the MSS demonstration model `demoOtterUSVPathFollowingHeadingControl`. The course and the toolbox agree on it entry for entry.

3. **Solving for the two thrusts.** Delete the zero row and $\mathbf{B}$ becomes 2×2 with determinant $-2y_p \neq 0$, so there is exactly one solution — no pseudo-inverse is needed, and no freedom is left over. Inverting it, or equally adding and subtracting the two remaining equations, gives

$$\begin{bmatrix} T_L \\ T_R \end{bmatrix} = \mathbf{B}^{-1} \begin{bmatrix} X \\ N \end{bmatrix} \qquad \Longrightarrow \qquad T_L = \frac{X}{2} + \frac{N}{2y_p}, \qquad T_R = \frac{X}{2} - \frac{N}{2y_p}$$

Read in words: **both propellers carry half the surge force, and the moment is made by adding to one and taking the same amount from the other.** The two tasks are superposed, and y_p is the lever that converts moment into thrust difference. A short lever costs more thrust for the same moment: $N = 47$ N·m over $2y_p = 0.79$ m needs 60 N of difference.

The `allocation` block writes this as the solve rather than as the two expressions, so that the code and this line say the same thing; the listing follows line 8.

4. **From thrust to shaft speed.** A propeller's thrust goes as the square of its shaft speed, with a different coefficient going astern because the blade is the wrong way round:

$$T = \begin{cases} k_{pos}\, n^2, & n \geq 0 \\ -k_{neg}\, n^2, & n < 0 \end{cases} \qquad \Longrightarrow \qquad n = \begin{cases} +\sqrt{T/k_{pos}}, & T \geq 0 \\ -\sqrt{-T/k_{neg}}, & T < 0 \end{cases}$$

This is where the linearity ends. Everything above was a linear sharing of two forces; the square root is not linear, so **equal steps in N are not equal steps in n.** Near zero thrust a small change in T moves n a great deal, and near full thrust it barely moves it.

5. **What the propellers can actually give.** Putting the shaft limits through the same curve gives the thrust each propeller can reach:

$$T_{\max} = k_{pos}\, n_{\max}^2 = 119.68 \text{ N}, \qquad T_{\min} = -k_{neg}\, n_{\min}^2 = -66.71 \text{ N}$$

Astern is weaker than ahead by almost a factor of two — the same blade, run backwards.

6. **The limit on the moment, and why it moves.** Both thrusts must lie in $[T_{\min}, T_{\max}]$. Substituting line 3 and solving each inequality for $|N|$:

$$\frac{X}{2} + \frac{|N|}{2y_p} \leq T_{\max} \implies |N| \leq 2y_p\left(T_{\max} - \frac{X}{2}\right), \qquad \frac{X}{2} - \frac{|N|}{2y_p} \geq T_{\min} \implies |N| \leq 2y_p\left(\frac{X}{2} - T_{\min}\right)$$

The smaller of the two is $N_{\lim}(X)$. The first branch falls as X grows — the harder the vessel is pushed, the less headroom the loaded propeller has. The second rises — a faster-turning inside propeller has further to fall before it reaches its astern limit. They cross at

$$X^\star = T_{\max} + T_{\min} = 52.97 \text{ N}, \qquad N_{\lim}(X^\star) = 73.62 \text{ N·m}$$

so **the turning authority of this hull is greatest at a moderate surge force, not at full power.** The Week 4 constant `N_max` is nothing more than this expression evaluated once, at $X = X_{ff} = 60$ N.

7. **What breaks when X stops being constant.** The Week 4 autopilot clamps its output with an ordinary Saturation block holding a fixed `N_max` computed from $X_{ff} = 60$ N. Leave that number in place and add a speed loop that can push to $X_{\max} = 120$ N, and line 6 says the true limit there is 47.15 N·m while the block still holds 70.85. The allocator then asks for $n_R = 116.2$ rad/s against a limit of 103.9 — **an impossible command, produced by arithmetic that was correct one week earlier.**

   The fix is not a gain and not a new block. It is to evaluate line 6 **at the largest surge force this model can ask for** rather than at the one it used to ask for. `W05_0_setup` therefore holds two numbers, and every model's Saturation names the one that belongs to it:

$$N_{\max} = N_{\lim}(X_{ff}) = 70.85 \text{ N·m}, \qquad N_{speed} = N_{\lim}(X_{\max}) = 47.15 \text{ N·m}$$

This is deliberately the conservative choice: while the speed loop is only asking for 77.6 N the true limit is 63.9 N·m, and the vessel is being held to 47.15. Line 3 of the reading table below measures what that costs. The alternative — recomputing $N_{\lim}$ from the live X at every step — buys back that margin at the price of four more blocks in the diagram, and is left as the last **What to try**.

8. **And the plant clamps anyway.** MSS `otter.m` limits n to $[n_{\min}, n_{\max}]$ inside itself (lines 167–170). An allocator that does not clamp therefore logs a shaft speed the vessel never turned, and every later reading of that log is wrong by however much was cut off. The allocator clamps, so the log records **what the plant received**.

The whole derivation, as one block

Lines 2 to 8 are a single MATLAB Function block. Below is the code it holds, copied out of `W05_H_speed_path.slx`; double-clicking `allocation` shows the same text. The comments are the block's own.

```
function n = allocation(N, X, y_pont, k_pos, k_neg, n_max, n_min)
%#codegen
%ALLOCATION  전진력 X 와 요 모멘트 N 을 두 축의 회전수 n = [n_L; n_R] 로.
%            The surge force X and the yaw moment N to the two shaft speeds.
%  X 는 **속도 루프의 출력**이다 (3주차). 상수가 아니라는 것 하나만 다르다.
%  X is the output of the speed loop of Week 3, not a constant. That is the
%  only difference; the two equations below are unchanged.

% 1) 두 힘을 두 추력으로. 배치행렬 B 를 세우고 **풀면** 된다 - 손으로 푼
%    T = X/2 +- N/(2 y_pont) 와 같은 것이고, 이 쪽이 5-7 절의 유도와 같은
%    모양이다. MSS 데모도 이렇게 한다: demoOtterUSVPathFollowingHeadingControl
%    의 Allocation 안에 Constant [1 1; 0.395 -0.395] 와 행렬 나눗셈이 있다.
%    B is square and nonsingular here, so the two forces are shared by solving
%    rather than by two hand-written expressions. The MSS demonstration model
%    does the same: its Allocation holds the constant [1 1; 0.395 -0.395] and
%    a matrix division.
B = [1        1;              % 전진력  X = T_L + T_R        / surge force
     y_pont  -y_pont];        % 요 모멘트 N = y_pont(T_L-T_R) / yaw moment
T = B \ [X; N];               % 좌현, 우현 / port, starboard

% 2) 프로펠러 곡선 T = k n|n| 을 n 에 대해 푼다 (앞 k_pos, 뒤 k_neg).
%    Solve the propeller curve T = k n|n| for n (k_pos ahead, k_neg astern).
n = zeros(2,1);
for i = 1:2
    if T(i) >= 0
        n(i) =  sqrt( T(i) / k_pos);
    else
        n(i) = -sqrt(-T(i) / k_neg);
    end
end

% 3) 프로펠러가 낼 수 있는 범위로 자른다. **otter.m 이 안에서 이미 자르기 때문에**
%    (MSS otter.m 167-170 행) 여기서 자르지 않으면 로그에 적히는 n 이 플랜트가
%    실제로 받은 n 과 달라진다 - 기록은 플랜트가 받은 것이어야 한다.
%    Clamp to what the propellers can do. otter.m clamps internally (lines
%    167-170), so without this the logged n would not be the n the plant got,
%    and a log must record what the plant received.
n(1) = min(max(n(1), n_min), n_max);
n(2) = min(max(n(2), n_min), n_max);
end
```

In the code	Which line of the derivation	Note
`B = [1 1; y_pont -y_pont];`	line 2, with the zero row already deleted	the matrix is built from `y_pont`, so a different hull needs no new code
`T = B \ [X; N];`	line 3	the backslash solves rather than inverts; for this 2×2 the two are identical, and $\det \mathbf{B} = -2y_p$ is why it may be used at all
`N` and `X` as the two input ports	line 7	**both are now signals**: `N` from the heading autopilot, `X` from the speed loop. Weeks 3 and 4 each had one of them as a constant
the `for i = 1:2` loop	line 4	one propeller at a time; neither can lend thrust to the other
`n_max`, `n_min` in the last two lines	line 8	the clamp that keeps the log equal to what `otter.m` received
nothing in the code	lines 5–7	N_{lim} is **not** computed here. It is evaluated once in `W05_0_setup` and sits in the autopilot's Saturation, which is why an out-of-date value there cannot be caught by the allocator

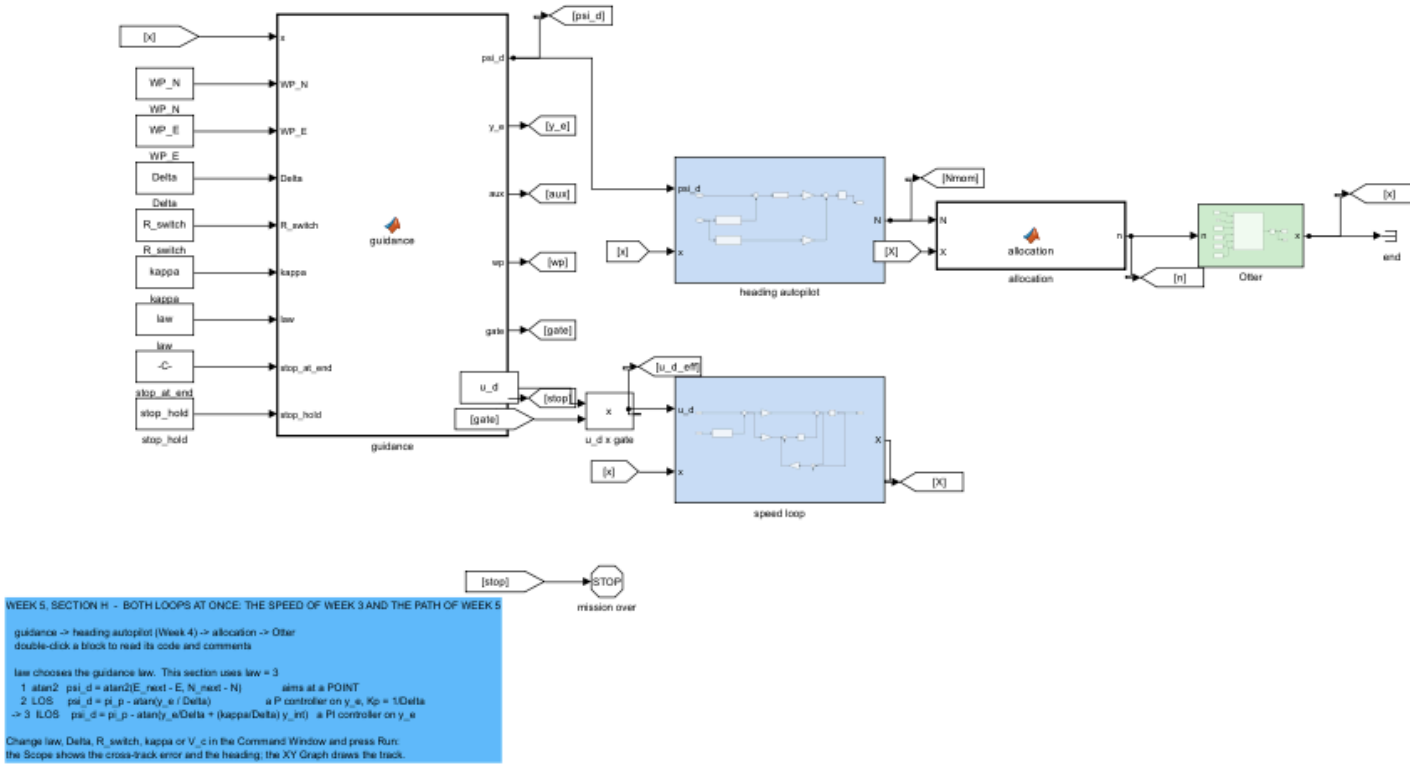
The last row is the whole point of §5-7: the block below the fault is correct, and the fault is still produced, because the number in the Saturation above it was computed for a surge force the model no longer uses.

> **The speed loop itself is not new**
>
> It is Week 3 unchanged, gains included: $X = K_{p,u}(u_d - u) + I$, $|X| \le X_{\max}$, with the integral that a drag-only axis needs and no derivative term. What §5-7 adds is not a controller but the accounting between two controllers that were designed separately.

Experiment 5-7 · Speed control and waypoint following together (15 min)

The same mission and the same ILOS law are run twice, with the heading autopilot of Week 4 untouched. The single change is where the surge force comes from.

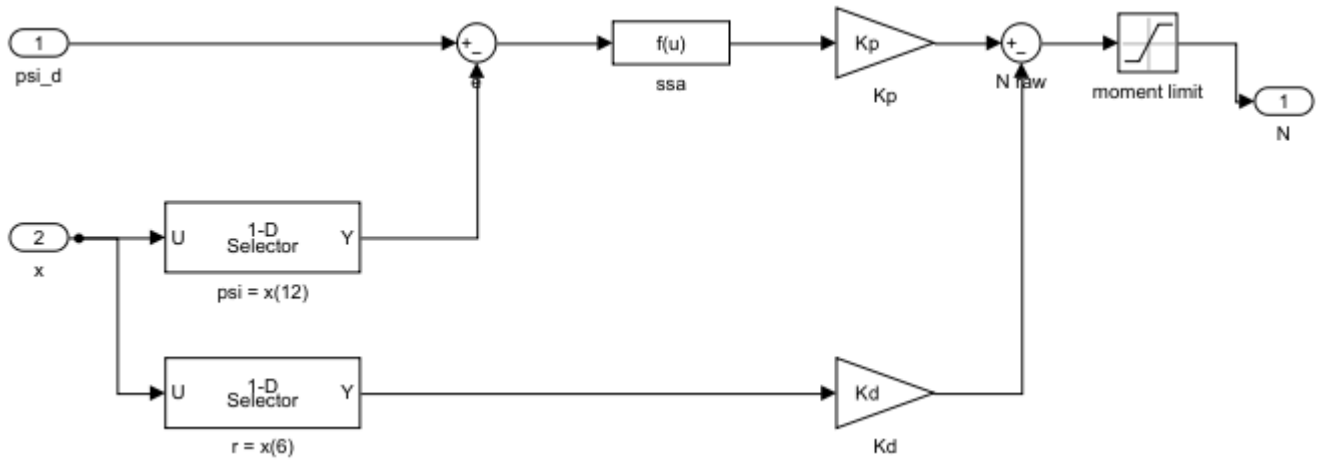
Model	Surge force	The Saturation in its autopilot holds
`W05_G_tuning`	the constant `X_ff` = 60 N, from a Constant block on the canvas	`N_max` = 70.85 N·m
`W05_H_speed_path`	the output of a `speed loop` subsystem, holding `u_d` = 1.0 m/s	`N_speed` = 47.15 N·m



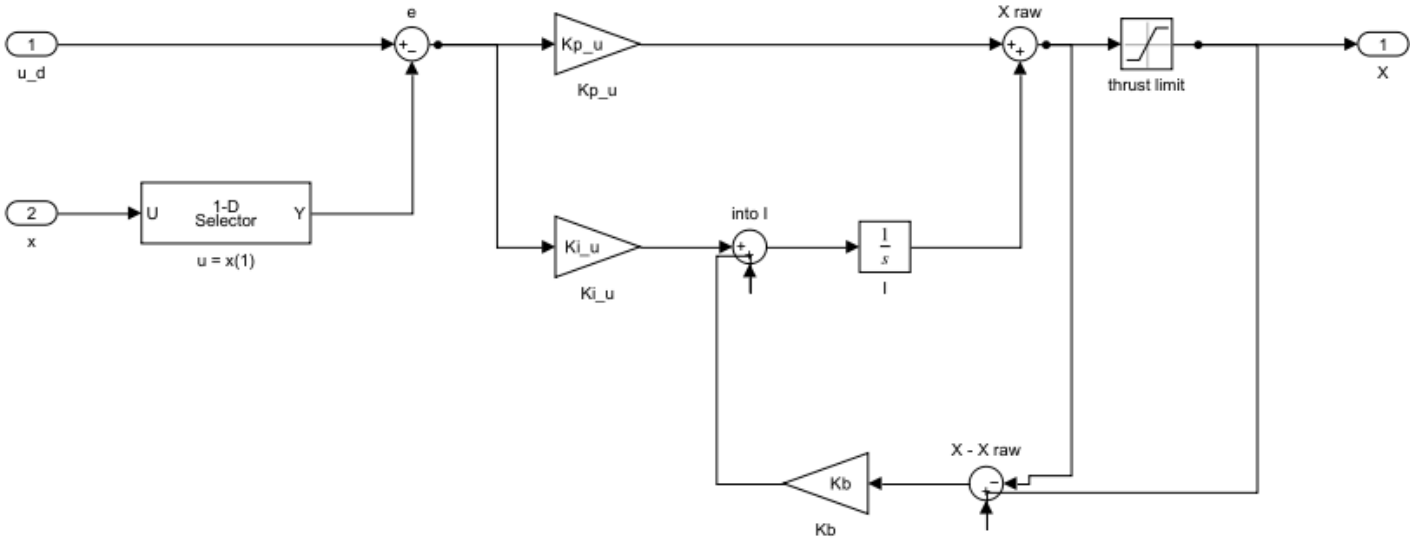
In the diagram	What it is
the six Constant blocks on the left	the guidance inputs, on the canvas rather than in a mask: `WP_N`, `WP_E`, `Delta`, `R_switch`, `kappa`
`guidance`	the ILOS law of §5-5, giving ψ_d and publishing y_e, the integral state and the leg index
`heading autopilot`	the Week 4 law, as a **subsystem of Simulink blocks**
`speed loop`	the Week 3 law, likewise. Its command `u_d` arrives from a Constant on this canvas
`allocation`	line 3 of the derivation, taking `N` and `X` and giving `n`
`Otter`	the hull. It receives exactly the `n` that appears in the log

Only the left half of the canvas is shown; the measurement, Scope, XY Graph, log and Animate blocks continue to the right.

Neither controller is a block of code. Opening either one shows the equation as a diagram, with the same blocks and the same names Weeks 3 and 4 used — one `Fcn` block for `ssa`, `Gain` blocks whose value is the name of a workspace variable, and an ordinary `Saturation`. Every gain is positive: **a term that is subtracted is subtracted at a summing junction**, so a sign appears in exactly one place on the diagram.



In the diagram	Which part of the Week 4 law
the two `Selector` blocks	$\psi = x(12)$ and $r = x(6)$, the only two states the law uses
the summing junction and `ssa`	$e = \mathrm{ssa}(\psi_d - \psi)$, wrapped by one `Fcn` holding `atan2(sin(u), cos(u))`
`Kp` and `Kd`	$K_p e$ and $K_d r$. **The D term is fed by the yaw-rate Selector, not by a derivative block** — there is no derivative block in the diagram
the `-` on `N raw`	where $K_d r$ is **subtracted**. The sign lives at the junction and nowhere else, so the diagram reads as the equation does: $K_p e$ minus $K_d r$
`moment limit`	the Saturation of line 7, holding `N_speed` here and `N_max` in the other six models



In the diagram	Which part of the Week 3 law
`u_d` on the left	the commanded speed, arriving as an ordinary signal from the canvas outside
`e`	$e = u_d - u$, with the command on the left edge and the measurement entering from below
`Kp_u`	the proportional term
`Ki_u` into `I`	the integral. `I` is an ordinary **continuous Integrator**, $1/s$ — the plant is continuous, and so is every controller in this course
`X raw` and `thrust limit`	$X_{raw} = K_{p,u} e + I$, then $\lvert X \rvert \le X_{\max}$
`X - X raw` into `Kb`, back into `into I`	the **anti-windup**, and the reason the loop has one

The anti-windup deserves its own line, because on a diagram it is the only path that runs right to left. It is **back-calculation**, the method of Week 3 §3-F, and the whole of it is one subtraction and one gain:

$$\dot{I} = K_{i,u}\, e \;+\; K_b \left(X - X_{raw} \right), \qquad X = \mathrm{sat}(X_{raw})$$

Symbol	Quantity	Value · source
X_{raw}	what the controller asked for, before the limit	N
X	what the propellers were actually given	N, $\lvert X \rvert \le X_{\max} = 120$
K_b	back-calculation gain	1 s^{-1}, the value Week 3 chose; `W05_0_setup`

While the force is inside its limit, $X - X_{raw} = 0$ and the term is not there at all — the loop is the plain PI of Week 3. The moment the limit bites, the term goes negative and **the integral is pushed back down by exactly the amount that was thrown away**, so it stops climbing towards a force that cannot be produced. Without it the integral keeps growing while the vessel accelerates, and then has to be unwound before the loop can respond at all, which is seen as an overshoot.

Setting `Kb = 0` removes it, and Experiment 5-7 runs both:

K_b	X at its limit for	speed overshoot
1	0.66 s	0.0216 m/s
0	0.92 s	0.0380 m/s

The difference is real but small, and the reason is worth saying plainly: **this mission barely saturates.** The surge force reaches 120 N only in the first second, while the vessel is accelerating from rest, and for the remaining 400 s it sits at 77.6 N. Anti-windup earns its place where a limit is held for a long time — the step of Week 3 §3-F, or a current strong enough to keep the force against its stop. It is carried here because it costs two blocks and because the case it protects against is not visible in advance.

> **Where the numbers live**
>
> Commands are Constant blocks on the canvas — `WP_N`, `WP_E`, `Delta`, `R_switch`, `kappa`, `u_d`. Gains are the values of `Gain` blocks, written as the variable's name. Both read the workspace, so `Kp = 400` in the Command Window followed by Run is the whole edit. The MSS demonstration model is arranged the same way: its waypoint lists, Δ, R and desired surge thrust are Constant blocks, and its autopilot gains are `Gain` blocks inside the `Heading autopilot` subsystem.

To produce every figure and number in this section

```
cd lectures/W05_simulink
W05_0_setup
W05_H_speed_and_path
```

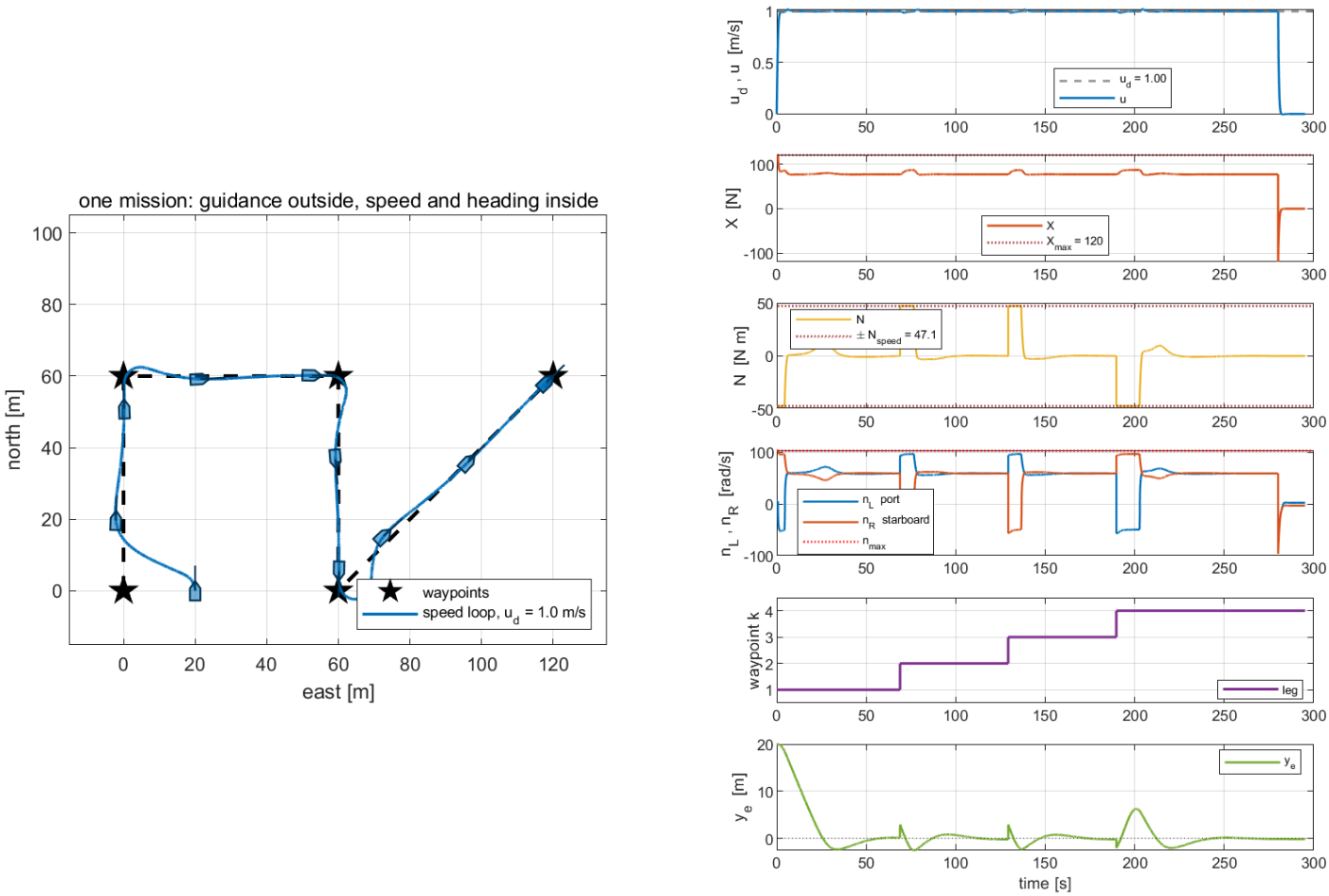
Actual output:

```
W05 Experiment 5-7  speed control and path following at once
  surge force from    u mean [m/s]   last leg at [s]   mean |y_e| legs 2-4 [m]   X mean [N]
  a constant X_ff        0.767             244.9                    0.353        60.00
  the speed loop         1.000             189.7                    0.896        78.88

  the allocation, checked by hand   (y_pont = 0.3950 m, k_pos = 0.01108)
    case              X [N]    N [N m]   T_L [N]   T_R [N]   n_L      n_R     n_L by hand
    straight leg      77.59      -0.50     38.16     39.43   58.683   59.651     58.686
    hardest turn     120.00     -47.15      0.32    119.68    5.357  103.931      5.357

  X [N]               0      20      40      60      80     100     120     140
  N_lim [N m]     52.70   60.60   68.50   70.85   62.95   55.05   47.15   39.25
  the constants the models hold:  N_max = N_lim(60) = 70.85,  N_speed = N_lim(120) = 47.15 N m

  the anti-windup (back-calculation), measured
    Kb      X at its limit for [s]   speed overshoot [m/s]
    1                    0.68               0.0216
    0                    0.94               0.0380
```



The figure draws the run that uses the speed loop and nothing else; the comparison with a constant surge force is the table above, where the two runs sit on two lines and can be read without a legend.

In the figure	Meaning
left	the mission, with the hull drawn along the track so the heading is visible at every mark
panel 1, u_d and u	the commanded speed and the one made — they lie on each other, which is what the loop was added for, and both fall to zero at 280 s when the mission ends
panel 2, X	the surge force the loop asks for: 77.6 N on a straight leg, up to $X_{\max}$ out of each corner
panel 3, N	the yaw moment, against the $\pm N_{speed}$ the Saturation holds
panel 4, n_L and n_R	the two shaft speeds, against $n_{\max}$
panel 5, waypoint k	the leg index, stepping up once per switch: $1 \to 2$ at 68.9 s, then 129.3 s, then 189.7 s
panel 6, y_e	the cross-track error against time: 20 m at the start, then a spike at each corner and a return to zero
every panel, after 280 s	flat: the vessel has arrived at the last waypoint and the run is being held for `stop_hold` seconds before it ends

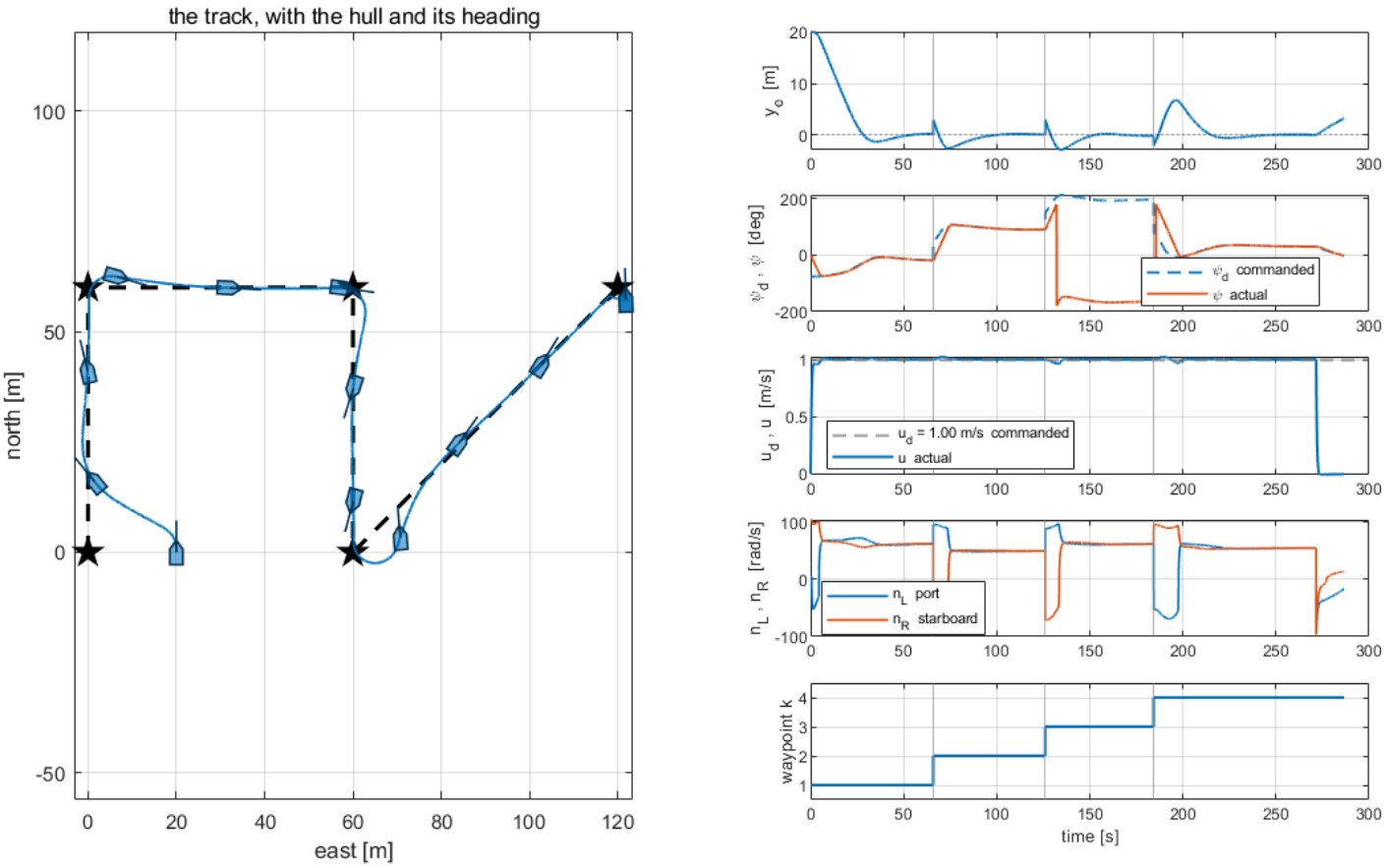
Reading the figure and the table against the derivation.

Where to look	What is there	Which line it settles
`u mean` column	0.767 against 1.000 m/s	line 1: a constant force gives the speed that balances drag, and 0.767 was never chosen by anyone
`X mean` column	60.00 against 78.88 N	the old models have no speed loop to vary the force; this one works for it
panel 2	77.59 N on a straight leg, rising to 120 N after each corner	holding 1.0 m/s costs 77.6 N of drag; a turn costs more, and the loop pays it until it reaches $X_{\max}$
`straight leg` row	$T_L = 38.16$, $T_R = 39.43$ N from $X = 77.59$, $N = -0.50$	line 3: $77.59/2 = 38.80$ each, then $\mp 0.50/0.79 = \mp 0.63$
`n_L by hand` column	58.686 against 58.683 logged	line 4 reproduced from the logged X and N alone, through the same square root
`hardest turn` row	$N = -47.15$ N·m exactly	line 7: the Saturation is holding `N_speed`, and the corner is asking for more than it
the same row	$T_R = 119.68$ N and $n_R = 103.931$	lines 5 and 7 together: at that moment the starboard propeller is **exactly at** $T_{\max}$, so its shaft is exactly at $n_{\max}$. The constant was chosen to make this the worst case, and it is
panel 4, at the corners	touching the red $n_{\max}$ line and never crossing it	the allocator can no longer ask for the impossible
panels 5 and 6 together	every step of the leg index is followed by a spike in y_e	§5-4: switching moves the path the vessel is steering to, so the error it measures jumps before the guidance works it down again
the $N_{\lim}$ row	70.85 at $X = 60$, 47.15 at $X = 120$	line 6: the first branch, falling as the surge force takes the headroom
the same row at $X = 0$	52.70 N·m, **below** the value at 60 N	line 6: the second branch, rising; the peak is between, at $X^\star = 52.97$ N
panel 3, the dotted pair	± 47.15 N·m, not the ± 70.85 the other six models hold	the price of the speed, stated as a number before any track is drawn
`mean	y_e	` column
`last leg at` column	189.7 against 244.9 s	23 % of the mission time, bought with the accuracy in the previous row

What the figure says

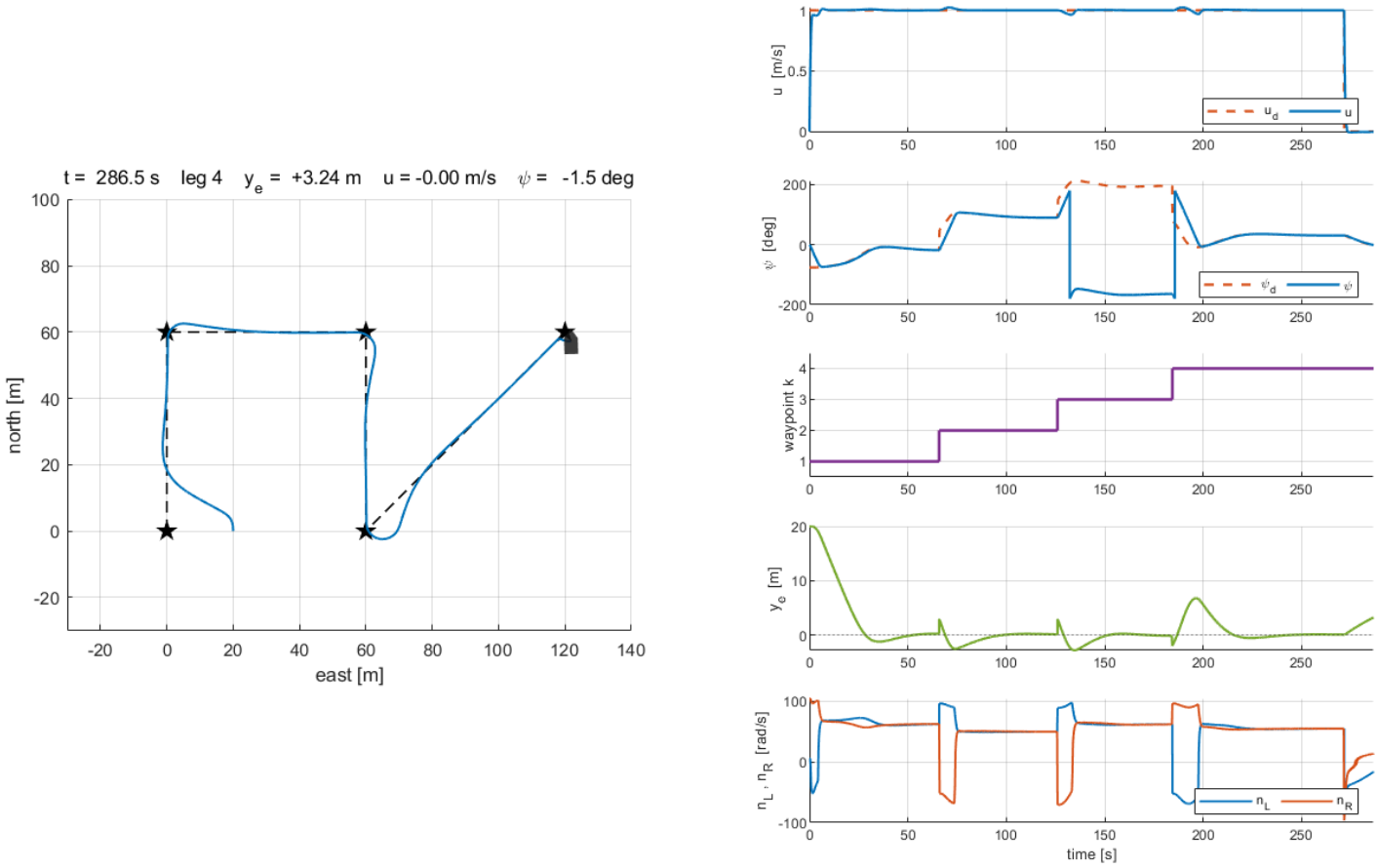
- Choosing the speed is possible, and it costs both moment headroom and following accuracy; neither cost is visible until the surge force stops being a constant.

Pressing Run on `W05_H_speed_path` alone produces the same numbers and a figure of the single run, because the model's `StopFcn` calls `W05_plot`:



In the figure	Meaning
left	the track with the hull drawn along it, and the five waypoints
y_e	the cross-track error, with a grey line at each switching instant
ψ_d, ψ	the commanded heading and the one achieved; the jumps to $\pm 180^\circ$ are the wrap of §5-2
u_d, u	the commanded speed and the actual one — they lie on top of each other, which is the point
n_L, n_R	the two shaft speeds, crossing over at every corner
waypoint k	the leg index, stepping $1 \to 2 \to 3 \to 4$ as §5-4's test becomes true

A second window opens **while the model is still running**, redrawn at every step by the `Animate` block, and it is the one to watch in class. Its left half is the mission as it happens, with the hull and its heading drawn at the vessel's position; its right half is five rows, each carrying one thing and each putting the command on top of what was achieved:



In the live view	What it answers
left, the hull on the track	where the vessel is **and which way it is pointing** — the difference between the two is the crab angle of §5-6
row 1, u_d and u	is the inner speed loop doing what it was told
row 2, ψ_d and ψ	is the inner heading loop doing what the guidance told it; the vertical jumps are the $\pm 180^\circ$ wrap of §5-2
row 3, waypoint k	which leg is active, and the instant each switch happened
row 4, y_e	the one number that says whether the guidance is working
row 5, n_L and n_R	what the two shafts are doing to produce all of the above

> ✎ **Switching the live view off**
>
> `animate = 0` in `W05_0_setup` leaves the block in place and makes it do nothing, which is faster when a script is sweeping gains. `pace = 1` runs the simulation at wall-clock speed so the window can be watched in real time; the numbers are identical either way.

★ The mission ends at the last waypoint

`W05_H_speed_path` stops when it arrives. The guidance applies **the same test it uses to switch legs** — less than `R_switch` left to the end of the leg — and, finding no leg to switch to, takes the speed command to zero through the `u_d x gate` product on the canvas. `stop_hold` seconds later a `Stop Simulation` block ends the run, which is why the figure above ends at 295.1 s rather than at `T_final` $= 400$ s. The vessel comes to rest at $(58.2,\ 118.1)$ m, 2.5 m from the last waypoint: `R_switch` is 3 m, and what remains is the distance covered while slowing down.

What ends it	Where it is
the test	inside `guidance`, step 3b: `arrived = near_end && k == n-1`
the speed going to zero	the `u_d x gate` Product **on the canvas**, so the end of the mission is visible in the diagram
the run ending	one `Stop Simulation` block, fed by the `stop` output
switching it off	`stop_at_end = 0` in `W05_0_setup`, which restores the old behaviour

The other six models keep `stop_at_end` unused: they hold the last leg and sail along its extension, which is what §5-4's reading table describes. A mission with waiting, re-planning and a state machine is Week 8; this is the smallest thing that deserves the name *end of mission*.

What to try	What to watch
`u_d = 1.6;` Run	the loop sits at $X_{\max} = 120$ N for most of the run, the corners are visibly wider, and the mission is barely faster — the drag has begun to win
`u_d = 0.4;` Run	slower, and `N_speed` is still 47.15: the vessel is being held to a limit it is nowhere near needing. The cost of a conservative constant, seen directly
`X_max = 70;` Run, after re-running `W05_0_setup`	`N_speed` rises to 66.40 N·m and the corners tighten, but the loop can no longer reach 1.0 m/s. The two are the same trade seen from the other end
`Kb = 0;` Run	the anti-windup path is switched off. The speed overshoot grows from 0.0216 to 0.0380 m/s — small here, because this mission holds the limit for well under a second
`animate = 1; pace = 1;` Run	the run takes as long as the mission does, and the live view draws the track, the error, both speeds and both shafts while it happens
Replace the Saturation in `heading autopilot` with a **Saturation Dynamic** fed by $N_{\text{lim}}(X)$, built from line 6	the margin between 47.15 and the true limit comes back, and the corners tighten. Measure `mean

✎ In class

- **Purpose** — show that two controllers designed a week apart share one piece of hardware, and that the sharing has arithmetic.
- **Point to** — the `hardest turn` row: $T_R = 119.68$ N, and $T_{\max} = 119.68$ N. It is exactly at the limit, by construction, because `N_speed` was chosen to put it there.
- **Ask** — "The vessel is pushed harder, so why can it turn *less* hard?" Both jobs come out of the same two propellers; surge takes headroom from yaw, and line 6 says how much.
- **Ask** — "Why is the turning authority greatest at 53 N and not at 0 N?" Below $X^\star$ the inside propeller runs out of astern thrust first, and astern is the weaker direction.
- **Ask** — "The Saturation holds 47.15 even while the true limit at $X = 77.6$ N is 63.9. Why not compute it live?" Because a constant is one block and a live limit is five, and the margin is thrown away only at the corners. Which of the two to build is an engineering question, not a matter of correctness — the last **What to try** turns it into a measurement.
- **Take away** — when a constant becomes a signal, every limit that was computed from it has to be computed again.

Part 2 · Laboratory run order

The experiments of Part 1 are worked through in order; this table is the index of what was run, for repeating the week at home.

Every example has **its own model**, laid out the same way: `guidance` → `heading autopilot` → `allocation` → `Otter`, each a commented MATLAB Function except the vessel. Every input those blocks take is a **Constant block on the canvas** — the waypoint lists, Δ, R, κ and the surge force are read off the diagram rather than hidden inside a block mask.

On the right of each model sit four things:

	What it shows	When
Scope	the cross-track error; the commanded and actual heading; the waypoint index stepping up	while it runs
XY Graph	the track, east against north	while it runs
Display	the active leg as a number	while it runs
Animate block	the live view: path, waypoints, track and hull on the left; y_e, u against u_d, n_L and n_R, and the leg index on the right	while it runs, when `animate = 1`

Setting `pace = 1` in the workspace runs the model **at wall-clock speed**, so the live view moves at the speed the vessel would; `pace = 0` runs it as fast as it can. The results are identical and only the waiting differs — a 10 s mission takes 12.7 s of wall clock at `pace = 1` and 1.4 s at `pace = 0`. When a run finishes, the model's `StopFcn` calls `W05_plot`, which prints the leg-by-leg table and draws the track with five traces beside it, so **pressing Run alone produces the same figure a section script does.**

Experiment	Model	Script	What it shows
5-0	all of them	`W05_0_setup`, `W05_1_build_guidance`	the parameters, and every model written from code
5-1	`W05_G_tuning`	—	where guidance sits, and the rotation that gives y_e
5-1b	`W05_G_tuning`	`W05_B_three_laws`	the `law` flag: one model run as atan2, LOS and ILOS
5-2	`W05_C_atan2`	`W05_C_aim_at_the_waypoint`	aiming at the next waypoint
5-3	`W05_D_LOS`	`W05_D_lookahead_distance`	LOS and the look-ahead distance
5-4	`W05_E_switching`	`W05_E_waypoint_switching`	the switching distance on a mission
5-5a	`W05_F_LOS`	`W05_F_LOS_in_a_current`	the offset a current leaves under LOS
5-5b	`W05_F_ILOS`	`W05_F_ILOS_removes_it`	the integral that removes it
5-5c	`W05_F_ILOS`, `law = 3` and `4`	`W05_F2_two_integrators`	the plain integral against the nonlinear one, at the same gain and at its own
5-6	`W05_G_tuning`	`W05_G_tuning_by_hand`	the tuning order on the mission in a current
5-7	`W05_H_speed_path`	`W05_H_speed_and_path`	the speed loop of Week 3 on the same hull, and how X and N split into n_L and n_R

- The scripts only repeat what Run already shows, for several values at once, and print the numbers quoted in Part 1.
- Each experiment ends with an **In class** note: purpose, what to point at, a question with its answer, and the sentence to take away.

Summary

Week Summary

Step	What was done	How it was verified
1	put a guidance law in front of the Week 4 autopilot	`check_overlaps` 0 in all seven models; y_e equals MSS `crosstrackWpt`
1b	put all three laws in one block behind a flag	Experiment 5-1b: one mission, $3.93 \rightarrow 1.52 \rightarrow 0.61$ m mean error for atan2, LOS, ILOS
2	atan2 against LOS	Experiment 5-2: 10.75 m against 0.04 m halfway to the waypoint
3	Δ as the P gain $1/\Delta$	Experiment 5-3: 118 s at 40 m, swinging at 0.5 m; $\Delta = 5$ m
4	the switching distance	Experiment 5-4: worst corner 2.98 m at $R = 3$ m
5	a current and ILOS	Experiments 5-5a and 5-5b: LOS offset $= \Delta \tan$(heading) to the millimetre; ILOS removes it
6	the tuning order on the mission	Experiment 5-6: $\kappa = 0.3$, legs 2–4 summed 0.92 m against 3.64 m for LOS
7	added the Week 3 speed loop to the same allocator	Experiment 5-7: 1.000 m/s held, last leg 23 % sooner; n_L by hand 58.686 against 58.683 logged
8	derived the moment limit the surge force leaves	Experiment 5-7: $N_{\text{lim}} = 70.85$ N·m at $X = 60$ N, 47.15 at 120 N, peak 73.62 at 52.97 N
9	drew both controllers as Simulink blocks, not code	`check_overlaps` 0 inside every subsystem; §5-2 to §5-6 reproduce every number of the MATLAB Function version unchanged

Progress Check

> ★ **Minimum condition for following Week 6**

Theory

- [ ] Able to compute y_e from two waypoints and a position.
- [ ] Able to explain why LOS is a P controller and which way Δ moves its gain.
- [ ] Able to predict the LOS offset in a cross current from the heading held.
- [ ] Able to explain what the integral of ILOS does and why its denominator limits windup.
- [ ] Able to write T_L and T_R from a surge force and a yaw moment, and convert them to shaft speeds.
- [ ] Able to explain why pushing the hull harder leaves less yaw moment, and where the moment limit peaks.

Laboratory

- [ ] `W05_1_build_guidance` ran and every model reported `overlapping lines: 0`.
- [ ] A parameter was changed in the Command Window and the XY Graph showed the new track.
- [ ] `W05_H_speed_path` was run with `animate = 1` and the live view drew the track, the error, both speeds and both shafts.
- [ ] `W05_check(1)`, `(2)` and `(3)` pass on a model built from `W05_P1_start`.

In-class laboratory — build the guidance layer by hand

The second hour of the Week 5 session is spent building the guidance block in Simulink. A complete Week 4 vessel is provided with one input left open: the commanded heading.

```
cd lectures/W05_simulink/problems
W05_P1_start                % creates W05_P1.slx — a Week 4 vessel, no guidance
W05_check(1)                % run this whenever, as often as needed
```

	Problem	Time	The number it must reproduce
1	The LOS law on leg 1	25 min	$\pi_p = 0$ exactly; y_e from 18 m to 0; $\psi_d \to \pi_p$
2	atan2 against LOS, same vessel	20 min	LOS holds the line; atan2 does not
3	A current, and the offset that stays	15 min	settled $y_e = \Delta \tan$ of the crab angle

- The problem sheet is `W05_simulink/problems/README.md`, and reference answers are in `W05_simulink/solutions/`.
- Problem 3 ends where §5-5 continues: the integral that removes the offset.

Assignment 5

- **Due**: before the Week 6 session
- **Submit**: the modified `W05_0_setup.m`, the numbers requested below, and two figures

① Requirements

Double the legs of the mission (every waypoint coordinate times two, legs of 120 m) and tune Δ, R and κ again by the order of §5-6 in a 0.3 m/s current.

② Verification — mandatory

> **★ A claim is not a result. Numbers are required, with the averaging window stated**

1. The table of Experiment 5-3 on one 240 m leg with the chosen Δ and its neighbours.
2. The table of Experiment 5-4 on the new mission.
3. The table of Experiment 5-6 on the new mission, and the κ chosen.
4. The LOS offset in the current, next to $\Delta \tan$ of the heading held.

③ Analysis (5–10 lines)

Compare the chosen κ with 0.3, and explain from the leg length why it moved in that direction.

Grading

Criterion	Weight
The model runs and produces the requested output	25%
Verification performed and numbers reported	40%
Correctness of the analysis	25%
Readability of the code	10%

Troubleshooting

Symptom	Cause	Fix
`W05_0_setup` stops in `mss_path`	the MSS toolbox is not at `Tools\MSS`	place MSS there; see the first page
the XY Graph shows nothing	the track is outside its axes	the axes are −20 to 140 m; a custom path needs `set_param(... '/track', 'xmax', ...)`
the vessel circles a waypoint	R smaller than the vessel can turn inside	raise `R_switch`
`guidance` complains that `WP_N` is undefined	`W05_0_setup` was not run	run it; the guidance block reads the waypoints from the workspace
a Scope does not match these notes	a changed variable is still in the workspace	run `W05_0_setup` again

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References

Primary

- Fossen, T. I. *Handbook of Marine Craft Hydrodynamics and Motion Control*, 2nd ed., Wiley, 2021, §12.3 (line-of-sight guidance and its integral form).
- Børhaug, E., Pavlov, A. and Pettersen, K. Y. "Integral LOS control for path following of underactuated marine surface vessels in the presence of constant ocean currents." *47th IEEE Conference on Decision and Control*, 2008, pp. 4984–4991. DOI: 10.1109/CDC.2008.4739352.
- MSS toolbox, `Tools/MSS/GNC/crosstrackWpt.m` and `ILOSpsi.m`.

In this course

- Week 2 §2-6 to §2-8 and §2-12: P, I and the tuning order. Week 4 §4-4: heading is not course.
- The previous, model-based version of this week — full derivations, Lyapunov stability and ALOS — is kept in `W05_simulink/_previous_version/`.

Next Week

Week 6 — Control Allocation. This week gave the vessel the square allocation of Week 4 and never questioned it. Week 6 asks what happens when there are more actuators than degrees of freedom, when a thruster saturates, and when one fails.